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ELECTRONIC ENGINEERING, TELECOMMUNICATIONS AND INFORMATION TECHNOLOGIES - APPLIED ELECTRONICS

Web Application for Trading Authenticated Sports Collectibles via Blockchain

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Mircea-Adrian PIELEANU

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List of Abbreviations

Abbreviation	Definition
ACID	Atomicity, Consistency, Isolation, Durability
API / APIs	Application Programming Interface(s)
CI	Continuous Integration
COA / COAs	Certificate(s) of Authenticity
CSRF	Cross-Site Request Forgery
DB	Database
Devnet	Development Network
DNA	Deoxyribonucleic Acid
eBay	eBay (online marketplace)
EU	European Union
FBI	Federal Bureau of Investigation
FR	Functional Requirement
GDPR	General Data Protection Regulation
HDD	Hard Disk Drive
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ID / IDs	Identifier(s)
JSA	James Spence Authentication
JSON	JavaScript Object Notation
JWT	JSON Web Token
KYC	Know Your Customer
MiCA	Markets in Crypto-Assets
NBA	National Basketball Association
NFC	Near-Field Communication

Abbreviation	Definition
NFT / NFTs	Non-Fungible Token(s)
NFR	Non-Functional Requirement
OPS	Operations
OS	Operating System
PDF	Portable Document Format
PSA	Professional Sports Authenticator
PSA/DNA	PSA Autograph Authentication
QR	Quick Response
REST / RESTful	Representational State Transfer
RPC	Remote Procedure Call
SDK / SDKs	Software Development Kit(s)
SEC	Securities and Exchange Commission
SHA-256	Secure Hash Algorithm – 256-bit variant
SQL	Structured Query Language
TLS	Transport Layer Security
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
US	United States
USD	United States Dollar
UX	User Experience
W3C	World Wide Web Consortium
WCAG	Web Content Accessibility Guidelines
XSS	Cross-Site Scripting

1. Introduction

THIS phenomenon of collecting sports memorabilia (from signed t-shirts and balls to rare athlete cards) represents a rapidly global market, valued at hundreds of billions of dollars. But this market has a big problem: how do you know if collector objects are real? Many estimates say that more than half of the most popular signed sports products might not be real. Because there are so many fake items, collectors and purchasers don't trust each other, which hurts both fans and the reputations of the clubs and athletes involved.

The risk of fake goods goes up as the market grows around the world. Dishonest sellers can sell phony signatures or artifacts that aren't what they say they are online. This hurts the reputation of the business and makes honest collectors less likely to trust it. So, making sure that each item is real is very important for keeping both fans and the history of sports safe.

Expert evaluations and certificates of authenticity (COA) that come as paper papers or holographic labels are what traditional authentication procedures use. These methods are useful, but they have certain problems. For example, paper certificates can be lost or falsified, and their validity typically depends on how trustworthy the issuing body is. Online markets have made it so that buyers and sellers are often far apart, making it hard to verify things in person. Blockchain is at the vanguard of new technologies being tested to make memorabilia authentication more secure and open. Blockchain technology makes it possible to store unique identifiers or "fingerprints" of goods on a ledger that can't be changed. This makes digital certificates that can't be changed and that anybody can check on their own. By keeping a permanent, unchangeable record of an item's legitimacy and ownership history on a public ledger, this method might get rid of any confusion about where it came from.

This thesis talks about OneFan, a web-based marketplace for trading verified sports collectibles. It uses blockchain to solve the problem of authenticity. OneFan's main new idea is to use a bespoke smart contract on the Solana blockchain to keep a unique cryptographic hash that shows that each collection is real. OneFan doesn't employ non-fungible tokens (NFTs) to show that someone owns a tangible thing, which is different from how most people do it. Instead, the platform uses a lightweight on-chain record: a cryptographic hash that acts as a digital certificate of validity and is kept safe on Solana. This is like an unchangeable fingerprint of the item's verified information, so any tampering or fraud would be clear on the ledger. OneFan stays away from the complexity, fees, and speculation that come with NFT trade by not minting full-fledged NFTs. Instead, it focuses only on authenticity and provenance. The blockchain record is not a coin that can be traded; it is a certificate that can be searched up to make sure that the platform verified a certain physical thing. In short, OneFan combines the trust of traditional expert verification with the security of blockchain recordkeeping to create a system where purchasers may trade memorabilia with confidence, knowing that cryptographic proof backs up the legitimacy of each item.

This is how the rest of the thesis is set up. In Section 2 (State of the Art), we look at the best solutions on the market for sports collectibles and the best ways to authenticate them using blockchain. We talk about their pros and cons. The third section, "Stakeholders," lists the main stakeholders of the OneFan platform, such as end-users and other parties with an interest in it. Section 4, "Statement of Work," lays out the project's goals, scope, and deliverables for making the OneFan app. Section 5 (Risks and Issues) looks at possible problems, such as technological, security, and market acceptance risks, and how to deal with them. Section 6, "Requirements Analysis," talks about the functional and non-functional requirements for customers, sellers, and administrators. Functional requirements include things like security, performance, and usability. Section 7 (Design) shows the system's design and architecture. It includes UML diagrams (use case and sequence diagrams) that show important scenarios including making a list, buying something, and interacting with the blockchain.Section 8 talks about how the application was put into use.Section 9 (Future Work) talks about possible improvements and additions to the platform that go beyond what it can do now. Finally, Section 10 (Conclusion) sums up what the initiative has done and what it means.

2. State of the art

THE trade of sports collectibles has traditionally been facilitated through marketplaces like auction houses, online platforms, and hobbyist forums. Today, big online shopping sites like eBay, as well as specialty sports auction sites like Goldin Auctions and Heritage Auctions, are important places to purchase and sell collectibles. New collector marketplaces are also starting to pop up. These platforms have made it easier for people all over the world to find collectibles, but they have also had to deal with the issue of authenticity directly. Marketplace owners have taken a number of steps in the past few years to make sure that purchasers get real things.

Using third-party authentication services is a typical method. For instance, eBay's "Authenticity Guarantee" program works with professional authenticators to check out some types of collectibles. When a valuable sports card or signed item is sold on eBay, it is sent to an independent authentication center first. There, experts check it out in person and make sure it matches the listing description. The item is only sent to the buyer after clearing this multi-point check. It usually comes with a new authenticated certificate or tamper-proof tag from the authenticators. In eBay's concept, this service doesn't cost the buyer or seller anything extra. Its goal is to get fake things off the market and build trust in the marketplace. Other platforms use similar methods. For example, professional sports memorabilia merchants often include paper COAs from well-known groups like PSA/DNA, JSA, and Beckett. Some even put holographic stickers on things that match an online verification number.

Even with these steps, the present solutions have problems. It takes a lot of work to rely on human expert verification, and it's not always right. Mistakes or even fake "authentications" have happened, and the process doesn't become easier as there are more transactions. In the end, buyers have to trust the expert or the middleman, and any break in that trust chain (such a dishonest verifier or a missing certificate) can hurt the system. Also, these services usually check an item when it is sold, but future buyers can't easily find out where it came from unless they look at the papers that came with it. It can be hard to verify something again if the certificate is lost or if someone doubts its authenticity years later.

Another problem with traditional marketplaces is that they don't have a single, tamper-proof record of who owns what and how real it is. A series of receipts or letters could show where something came from, but these can be faked or taken away from the object. For example, a signed jersey that is worth a lot of money might come with a printed COA. If the COA is copied or switched, a less honest vendor could try to sell phony jerseys with copied certifications. It is still hard to keep track of collectibles as they change hands. Many collectors utilize reputation systems (such seller ratings and buyer feedback) to figure out if something is trustworthy. These methods are helpful, but they don't solve the main problem of checking the item itself.

In short, current marketplace solutions are changing to reduce fraud, such as eBay's authenticity program and specialized auctioneers that check items in-house. However, they still depend on centralized middlemen and physical certifications. This opens the door for software solutions that can decentralize and automate the process of verifying authenticity, which will make it less dependent on experts and more open for everyone in the market.

Blockchain technology is becoming a useful tool for making the collectibles market, especially sports memorabilia, more trustworthy and open. A blockchain can be used as a distributed ledger to keep track of a digital version of a real-world object, which is typically called a digital token or certificate. Once authenticity data on an object is published on-chain, it becomes very hard to change or fake because blockchains are databases that can't be changed and can be checked by anybody. This property directly deals with the problems of provenance and forgery that are common in the collectibles market.

Using Non-Fungible Tokens (NFTs) as digital twins of real-world assets is a common way to authenticate on the blockchain. An NFT is a one-of-a-kind token on a blockchain, like Ethereum or Solana, that can hold metadata about a certain item. Sellers make a permanent digital certificate that can be transferred with the item by minting an NFT that is linked to a tangible collectible. For instance, Chiliz, a firm that makes sports fan tokens, started GameUsed.com, an authentication service that verifies major sports memorabilia (such game-worn jerseys or championship game tickets) and then mints them as NFTs on the Chiliz blockchain. In their idea, each physical thing has a cryptographic NFC chip (Near Field Communication tag) attached to it. Scanning the chip lets you get the item's NFT and metadata from the blockchain. The NFT shows that the item is real and that the seller owns it. If the NFT is real and still linked to the seller, the item is probably real too. This does a good job of connecting the actual world and the digital world: "An NFC chip connects physical items with the NFT on-chain that goes with them." By 2022, sports NFT projects like NBA Top Shot (for digital video highlights) showed that digital artifacts could be sold for a lot of money, with sales of over $1 billion. Building on this, tokenizing physical memorabilia through NFTs is thought to be the next big thing, with estimates that the market for NFT sports collectibles may reach $92 billion by 2032.

Digital product passports or certificates are another form of blockchain-based authentication that don't require the thing to be tokenized for trade. For example, the platform collectID puts secure NFC tags on collectibles and then uses blockchain to keep track of each tag's unique identification. When someone taps the object with their phone, the system checks the tag's cryptographic signature against the blockchain record to make sure it is real right away. CollectID makes a digital record (which might be an NFT or just a data entry on-chain) that connects the physical item to a blockchain entry. This "physical-to-digital link" method makes sure that each item of memorabilia has a unique, unforgeable digital identity on the blockchain. This makes transactions far more trustworthy. In these kinds of systems, the blockchain entry usually has or points to the item's data (including its type, serial number, origin, and verification date) hashed or encoded in a way that would make it clear if the physical tag was tampered with or copied because it wouldn't match the blockchain record.

Blockchain authentication methods have a number of benefits over more traditional ways: (1) Immutability: Once authenticity is recorded, it can't be changed or removed. This means that

any future buyer can check the historical record of the item's authenticity and ownership. (2) Decentralized trust: Verification doesn't depend on trusting a single company's database or paper trail, but rather on a consensus-secured public ledger. (3) Ease of transfer: Digital certificates on blockchain can be easily transferred to new owners when an item is sold, keeping the provenance intact. This stops situations where missing documents break the chain of custody for an item. Also, connecting actual objects to the blockchain can help stop theft and fraud in secondary markets. For example, if an item is stolen or not verified, it won't have a matching blockchain certificate, which will let purchasers know that there might be a problem. Sports groups also support these kinds of regulations since protecting authenticity helps keep their brand strong and keeps fans safe from scams.

But the blockchain solutions that are already out there frequently have their own problems and trade-offs. Early NFT-based marketplaces, especially on Ethereum, had large transaction fees and slow throughput. This made it expensive to manufacture or move tokens for each low-value item. Also, adding NFTs can make things harder for consumers because buyers and sellers need digital wallets and some knowledge of cryptocurrencies, which can be a problem for people who aren't tech-savvy. Some collectors are also apprehensive about mixing tangible collectibles with digital assets that could be exchanged on unstable crypto marketplaces because NFTs are so speculative. There is also the challenge of the physical-digital link: how to make sure that a physical thing is linked to its digital token. NFC tags, QR codes, and serial-numbered holograms are some of the solutions that are employed, but they all need to be tamper-proof to work. Taking off or copying the tag could break the system. So, blockchain is the backbone of a stronger authenticity system, but the details of how it will be used (which blockchain to use, whether to use tokens or data, and how to connect it to tangible goods) are very important for its success and acceptance.

How OneFan Works: When it comes to OneFan, the platform uses blockchain in a smart way to get the most benefits and the least amount of problems. OneFan doesn't give out NFTs for each item, which would let people trade them without the physical object. Instead, they employ a special smart contract on Solana to store a cryptographic hash for each verified collectible. This method creates an unchangeable certificate of authenticity on-chain without making a token that may be moved. The hash is made from the object's authenticity data, which might be a JSON file with the item's ID, description, and the verifier's credentials, or a digital certificate file. This makes sure that the hash is just for that item. It doesn't have any market value, so people can't trade it for profit. Its only purpose is to prove that it is real. This method is similar to the "digital fingerprint" idea that has been written about: the hash is like the item's fingerprint on the blockchain. Anyone with the item's reference, like an item ID or QR code from OneFan, can use the Solana blockchain to get the stored hash and check that it matches the item's expected data, which proves that it is real. OneFan doesn't use NFTs, which means they don't have to deal with complicated things like transferring token ownership or paying listing fees on marketplaces. Instead, they only focus on keeping the authenticity record safe.

In short, blockchain-based verification for sports collectibles is a cutting-edge solution that is always becoming better. These solutions promise a new level of confidence, from full NFT tokenization of collectibles to simplified on-chain certificates. OneFan builds on these ideas by using a streamlined Solana-based certificate system that combines the security of blockchain

verification with the ease of use of a regular marketplace. This combination of blockchain trust and usability is meant to get beyond the problems with both the old paper-based systems and the newer NFT-based ones. In addition to technological issues, blockchain-backed authenticity will only become widely used if there are clear rules and if the user experience is improved. Governments are starting to write rules for digital assets that are related to laws that protect consumers, stop money laundering, and secure personal data. For example, Europe's MiCA framework and the U.S. SEC's changing advice on digital collectibles. Because of this, marketplaces need to make sure that their on-chain certificates don't accidentally turn into regulated securities. At the same time, they need to make sure that personal data, like customer names and shipping information, stays off-chain or is cryptographically hidden to follow privacy laws like GDPR. Projects that can find this balance will have an easier time following the rules and will make mainstream collectors feel more confident.

Frictionless wallet management is becoming a key feature that sets wallets apart in terms of usability. There are currently several new custodial services and "wallet-as-a-service" APIs that let platforms make a safe key pair for each user in the background. These keys are safeguarded by familiar login information instead of complicated seed phrases. When used with chains that have very low fees, like Solana or Layer-2 rollups on Ethereum, this invisible-wallet method makes it almost impossible for non-technical buyers and sellers to understand how to use it. Lightweight mobile SDKs also let you do things like tap-to-verify NFC checks right from your phone, which adds the same degree of trust to in-person deals at card shows or sports memorabilia conventions.

Lastly, more and more businesses are trying out hybrid provenance models. In these models, high-resolution images and notarized appraisal documents are stored off-chain (to keep the blockchain from getting too big), and a single Merkle-root hash is anchored on-chain. This combines the auditability of distributed ledgers with the scalability of traditional cloud storage. These new ideas suggest that the future generation of platforms, including OneFan, will be able to provide collectors the best of both worlds: the security and decentralization of blockchain certification and the speed, low cost, and ease of use of traditional online shopping experiences. As these enabling technologies improve, blockchain authentication is likely to go from being a specialized feature to an industry-wide standard for protecting the value of sports collectibles.

3. Stakeholders

IMPLEMENTING the OneFan platform involves understanding the needs and concerns of various stakeholders who have an interest in the system. The primary stakeholders and their roles are outlined below:

- Collectors/Buyers: These are people who are sports enthusiasts, collectors, or investors who buy memorabilia through the OneFan platform. Their major purpose is to buy real collectibles with trust that they are real. They want a way to check that something is real, a smooth buying experience, reasonable prices, and protection against scammers. Buyers are also worried about the safety of their transactions and personal information. With OneFan, customers may use the web app to look through postings, check the authenticity of items (using the blockchain certificate lookup), buy things, and maybe even sell them again. For the platform to be successful, it is very important that purchasers have a good experience (simple search, clear verification of validity, and a way to settle disputes if there are any).
- Sellers: These are people or groups (such hobby shops, memorabilia dealers, or even sports clubs) that put sports collectibles up for sale on OneFan. Sellers desire a marketplace that is trustworthy and will give them a fair price for their goods. One of the most important things for sellers is how easy it is to market items, which includes giving proof of authenticity or provenance. Sellers on OneFan have to send in their collectibles for authentication. This means they have to give the platform information about the items and maybe proof (such photos or existing COAs). The platform's legitimacy helps them; if OneFan's verification makes buyers trust them more, sellers can sell more easily and maybe for more money. Sellers also need safety, though. For example, they need to know that if they send a real item, the sale will go through (to avoid false chargebacks) and that the platform's fees or rules are fair. They have a say in any rules about shipment, costs, and how to handle disputes.
- Platform Administrators (Admins): The OneFan admins are in charge of running the marketplace. They are in charge of a lot of things, like managing user accounts and roles, reviewing and approving new listings (which includes setting up or checking the validity of each item), and keeping the platform safe and compliant. Admins make sure that the regulations are followed. For example, they make sure that no illegal things are posted and that each item goes through the right verification process before it is put up for sale. They are also in charge of putting the authenticity hash on the blockchain when an item has been confirmed, and they are the ones who update the Solana smart contract. Admins will also help users with problems, settle disputes between buyers and sellers (for example, if a customer says an item they received is not as advertised), and generally keep the marketplace in good shape to retain quality and confidence. Their

job is to make sure the platform runs smoothly, keep its good name, and make sure that the blockchain integration and other systems work properly.

- Partners/Experts: OneFan doesn't need an outside company to issue NFTs, but it can still need experts or an internal team to check things before they can be listed. These could be expert authenticators who check goods (either in person or with high-resolution photos) to make sure they are real or have been used in a game. OneFan might work with sports groups or trusted authenticators for some kinds of collectibles. For example, it might work with a card grading agency for trading cards. These stakeholders are concerned about the correctness of the verification process and keeping their professional reputation. They interact with the admin workflow, maybe by giving their opinion, which the admin then utilizes to decide whether or not to approve. Part of stakeholder management here is making sure there is a clear way for experts to give their feedback (and maybe paying them or giving them credit in the item listing).
- Users (General Community): In addition to active buyers and sellers, there may be people in the community who interact with the platform's content in other ways, such looking through collections, following their favorite sellers, or using the site's resources like blogs and tutorials on how to authenticate memorabilia. They care about how useful the platform is for learning and may want to buy or sell things in the future. They want the platform to create a community of trust and fans.
- Development and Maintenance Team: Made up of the project supervisor and anyone else who helped build the OneFan system. From a project point of view, this set of stakeholders is worried about getting the software product to work, be safe, and be easy to maintain before the deadline (defense date). They have a say in picking the right technologies (like Spring Boot and Solana) that suit the needs and can be used with the resources they have. The development team also needs to think about who will be in charge of the project in the future. They need to provide solid documentation and tests and follow best practices so that the platform may be expanded or turned into a product after it is first deployed.
- Sports Clubs and Brands (Secondary Stakeholder): Sports clubs, leagues, or equipment manufacturers are not directly involved in the platform's activities, but they do have an indirect stake. If OneFan takes off, it can help official gear sales and safeguard brand value by being able to tell real items from fakes. There is a chance for people to work together (for example, a club may use OneFan to sell limited-edition items with blockchain certifications). These stakeholders value the fact that there are fewer fake collectibles on the market and that fans are more involved with real collectibles. They are interested in making sure that the law is followed (for example, not helping to sell stolen goods and respecting intellectual property).

The design and functionalities of OneFan were affected by knowing these stakeholders and what they wanted. The platform is designed to meet the needs of both buyers and sellers. It gives them a marketplace that is easy to use and builds trust. It also gives admins the tools they need to administer and protect the system. The authentication procedure is safe, which makes external partners and the community happy.

4. Statement of Work

THE main purpose of the OneFan project is to create, test, and prove a prototype web platform for trading sports collectibles that are guaranteed to be real. The platform uses blockchain technology, namely the Solana blockchain, to verify the authenticity of each item. This lets buyers and sellers trade memorabilia with more confidence. A full-stack application with a Spring Boot backend (including a database and business logic) and a web-based frontend interface is in scope. This design will support different user roles, such as Buyers, Sellers, and Administrators. Each position will have its own set of features that work together to make the marketplace safe and open. In real life, the OneFan prototype would let users manage their accounts (register, log in, and assign roles as buyer, vendor, or admin) and do basic marketplace tasks. Sellers can make new collectible listings by filling out information about the item, attaching pictures, and giving any proof of authenticity that is required. After that, the listings will go through an admin approval process. An administrator will have to look over each listing, evaluate the details (like an expert authenticity check), and then either approve or deny the item. When an item is approved, the system creates a digital certificate of authenticity in the form of a cryptographic hash. This hash is subsequently written to the Solana blockchain for permanent record-keeping. Buyers can see approved ads and browse and search through the things that are offered. They can also see the specifics of the items and their blockchain-backed authenticity certificate, and then they can buy the items using the platform's transaction procedure. A simple order management system will keep track of purchases that have been made. The goal of this project is to show that a blockchain-integrated marketplace can enable the entire process of listing, verifying, and exchanging sports collectibles in a single prototype system.

By the end of the project, the following deliverables are expected:

- Functional Web Application: A completely working web application (installed on a local development server or test environment) that covers all of the primary use cases, such as registering and logging in as a user, creating a list, getting admin approval, and buying an item. The Spring Boot backend, the web frontend, and how they work together are all part of this.
- Blockchain Smart Contract: That keeps track of authenticity data on-chain is called a Blockchain Smart Contract. This will be put on a Solana Devnet (a test network) or a Solana emulator that is running on your own computer. It offers a feature that lets you record a collectible's authenticity hash and get it back to check it.
- Database Schema: A organized database, like a SQL schema, that keeps all the information it needs, such as user accounts, collectible listings, transaction records, and links to authenticity certificates. This schema is the backbone of the backend and makes sure that data stays the same between sessions.

- Project Documentation: Full documentation for the prototype. This includes the thesis report, any user manuals or a README with instructions for running the system in a local or test environment, and inline code documentation that follows best practices for making the code easy to read and maintain. I also give design artifacts like UML diagrams (use case diagrams, class diagrams, and sequence diagrams that show how the system works and how it looks) to help others understand the system's design.
- Test Suite: A group of tests that check important parts of the program. This comprises unit tests for crucial backend parts, like functions that handle blockchain interactions or business logic, and integration tests for important end-to-end scenarios, like making a listing and buying something. The test suite makes ensuring that the prototype's features perform as they should and helps catch regressions throughout development.

Because OneFan is being made as a prototype in an academic setting, several features and things to think about are not included in this project on purpose. The main goal is to show off the essential capabilities, not to make a full-featured, production-ready solution. There are certain important limits that apply:

- Limited User Scale: The system isn't meant to handle a lot of users or transactions. The goal is to have a few hundred users and listings for demonstration purposes, not millions of users or a lot of transactions. It is not possible to scale to production-level yet (for example, optimizing for thousands of processes per second).
- No Mobile Application: The only way to develop is through a web-based interface. This prototype does not have a separate mobile app for Android or iOS or a design that works well on mobile devices.
- No Live Auctions or Advanced Marketplace Features: There are no complicated marketplace features like live bidding and auctions, real-time communication between buyers and sellers, or dynamic pricing systems. The prototype only works with fixed-price listings and simple buy/sell transactions.
- Payment Integration Placeholder: There is no integration with real-world payment gateways (to handle real money transactions). In the prototype, purchases are either simulated or handled with placeholder code instead of processing real funds. This focus shows that the project's main goal is the process of verifying authenticity, not executing money transactions.
- Use of Test Blockchain Network: All blockchain activities take place on the Solana Devnet (a public test network) or a local test environment, therefore no real coin is traded. The blockchain part is only used to store and retrieve authenticity data at a low cost. The prototype doesn't include handling actual Solana tokens or gas fees beyond the little quantities on a test network.
- Omitted Regulatory/Logistics Features: This academic prototype doesn't talk about critical real-world issues like KYC (Know Your Customer identity verification), taxes, shipping and logistics integration for delivering items, or following the law. If the

system were to grow beyond the proof-of-concept level, these would be areas that may be worked on in the future.

By clearly stating what is not part of the project's scope, the development stays on track and possible. The limits above make sure that the team stays focused on building the core authentication workflow and marketplace features without becoming distracted by extra features.

The project's development followed a normal software engineering lifecycle over the course of the school year, including phases for research, design, implementation, and testing. In the first step, a lot of research and requirement gathering were done to figure out what the sports collectibles market needed and what problems would arise when integrating blockchain (this included looking at current solutions and setting precise functional requirements for OneFan). Based on what I learned, I produced an architectural design and project plan that showed how the full-stack components (frontend, backend, and blockchain features) would work together. The implementation phase was done in steps, like how Agile works. Development was split into smaller parts, which made it possible to plan, code, and review features in cycles. The backend was constructed with Spring Boot (Java) for strong server-side functionality and RESTful API endpoints. A basic web client was used as the frontend for users to interact with. A set of features was created and tested with each iteration. For example, user administration in one iteration, listing generation in the next, blockchain certificate integration in the next, and so on.

I met with the project supervisor on a regular basis during this time to make sure we met academic standards and to quickly deal with any technological problems that came up. For example, early integration tests were done to make sure that the app could send and receive messages from the Solana Devnet (by writing and reading a test authenticity record) before adding more complicated capabilities. As far as timetable milestones go, the main features of the marketplace were supposed to be done by about the midway of the project timeline, which was at the end of the first semester or by spring. This included basic user accounts, listing and buying flows, but not the entire blockchain connection. The last few months of the development cycle were spent improving the blockchain part. This included adding the smart contract, connecting it to the backend, and making sure that the admin approval process works properly with the Solana network. During this time, the focus was also on addressing issues, improving the application, and doing comprehensive testing. Testing was a continual process. During development, unit tests were built for important modules like the blockchain interface service. After all the main features were in place, there was a final integration testing phase. The end result of this tiered, iterative process is a working prototype that was delivered on time for the thesis defense after going through multiple rounds of improvement and testing.

The OneFan project's success is based on its ability to show a full, real-world usage scenario on the prototype system from beginning to end. If a user can do all of the following things in the deployed demonstration environment, the platform will be seen as successful:

1. Account Registration and Role Assignment: A new user can sign up for an account on the site and then log in. You can give the user a role, such signing up as a Seller to list an item.

2. Listing Creation by Seller: The Seller user makes a new collectible listing by giving all the necessary information, such as a description of the item and pictures. When you submit the item, its status changes to "pending approval," and purchasers can't see the listing yet.
3. Admin Review and Blockchain Certification: An Administrator user logs into the system, looks at the pending listing, and checks the information that has been given. The admin then does an approval action, which is like authenticating the item. When you do this, the system makes a unique authenticity hash for the item's data and saves it on the Solana blockchain .Once this on-chain record is made, the listing is authorizes. This changes the item's status to "approved" and makes it available in the marketplace.
4. Buyer Browsing and Purchase: A Buyer user can now see the newly approved collectible listing when they look through or search the marketplace. The buyer looks at the item's details page, which offers information regarding the blockchain authenticity certificate, including a transaction ID or hash that can be checked on the Solana explorer to make sure that an authenticity record exists. The buyer is happy that the item is real and goes forward with the purchase through the platform's purchasing mechanism. For the prototype, this may be a simulated checkout since real payment processing isn't built in yet.
5. Order Confirmation and Record Keeping: When someone buys something, the system saves the transaction in its database and sends the buyer an order confirmation. The object is recorded as sold, and the chain of custody is updated. The ownership transfer could be indicated, however full ownership tracking on-chain may be easier in the prototype. At this point, the whole process, from putting a collection up for sale to verifying it on-chain and then selling it to a buyer, has gone off without a hitch.

The fact that the prototype can do this whole process without any problems shows that the OneFan idea is possible and works. Along with this functional scenario, success is also tested against important non-functional needs. Security is the most important thing: the system must make sure that no one can do anything without permission (for example, only admins can approve listings and users can only change their own data). This protects the marketplace's integrity. I also think about performance and dependability. Pages should load quickly, and transactions with the blockchain (such writing or reading the authenticity certificate) should happen rapidly enough that they don't get in the way of the user experience. The project will be considered a success if the prototype can run the end-to-end scenario above while also meeting basic security and performance benchmarks.

To sum up, this Statement of Work lays out a clear approach and scope for building the OneFan platform as a proof-of-concept system. It sets the project's goals and features, lists the expected deliverables, and makes clear what will and won't be included in the implementation. The timeline and method show how the project will be carried out in stages, and the success criteria provide specific targets for the finished prototype. The OneFan project's Statement of Work lays out all the details ahead of time, giving the team and stakeholders a clear plan for how to move forward. This makes sure that everyone is on the same page about the project's goals.

5. Risks and Issues

EVERY software project faces potential risks that could impact its success. For OneFan, which combines a web marketplace with blockchain technology, it was very important to find and fix major hazards as soon as possible. This section talks about the main risks and problems that are expected to happen, as well as how likely they are to happen, how bad they could be, and how to deal with them:

1. R1: Blockchain Integration Risk:
    - Risk: The chosen blockchain (Solana) or the custom smart contract may not work as planned, which could put the platform's ability to record and verify authenticity at risk. There could be problems with the smart contract code, problems connecting the Spring Boot backend to the Solana network, or problems with the Solana network itself. For instance, if the Solana network goes down or gets too busy, approval operations might not work, and items wouldn't have their authenticity hash recorded on-chain, which would go against OneFan's main value proposition.
    - Likelihood: Moderate (there is some risk involved with using several technologies together).
    - Impact: High, given the platform's main goal is to make sure that things are real.
    - Mitigation: To lessen the impact, the project leveraged well-documented libraries for blockchain interface, such as the Java-based Solana SDK (Solanaj), to handle remote procedure calls and transaction signing instead of building low-level code from scratch.In the early stages of development, a rudimentary version of the blockchain write/read cycle was made to test the idea (for example, writing a test hash to Solana Devnet and receiving it back).This early testing found problems with integration before the project was fully developed. The architecture also limits blockchain calls to a certain service module. If a blockchain transaction fails because of network problems, the system won't mark the item as fully approved, which keeps things from being inconsistent. Admins are told that the attempt failed so they can try again when the network is stable. In the worst-case scenario, a backup plan was made to temporarily store authenticity data off-chain (in the database) while correcting blockchain problems. This would keep the marketplace accessible (though without on-chain verification) until reliability was restored.
2. R2: User Key Management and Adoption of Blockchain:

- Risk: Adding blockchain to the mix makes people wonder how much complexity end-users would have to deal with. It could be hard for collectors who don't know anything about crypto to understand or use if they had to handle cryptographic keys or Solana wallets themselves. They might forget their keys or not want to pay fees to get cryptocurrencies, which would slow down adoption.
- Likelihood: Many sports memorabilia collectors don't know much about crypto wallets, so it's likely.
- Impact: Medium to High, because forcing people to know about blockchain could make it much harder for them to use.
- Mitigation: OneFan's design hides all the complicated parts of blockchain from the user on purpose. People who buy and sell things on OneFan don't have to make or operate Solana wallets at all. The site takes care of blockchain interactions for them in the background. The backend server actually has a secure Solana account (private key) that it uses to send authenticity transactions. Users don't have to worry about key management because they don't have to memorize seed phrases or deal with crypto tokens for gas fees. In fact, OneFan may pay for writing to the blockchain because Solana's transaction fees are so minimal (a few cents). This means that customers don't even know about "gas" or fees. The platform works well as a middleman between users and the blockchain network. This technique makes the user experience much easier (the blockchain part is hidden yet helpful), but it does mean that the server has to take care of the key (see R3). I planned to back up the server's key regularly, and the smart contract was set up so that even if the server's key were lost, the authenticity records that are already on-chain would still be publicly verifiable and linked to the items (the data on-chain is unchangeable and doesn't need to be used with the same key).

3. R3: Security Vulnerabilities (Web Application):

    - Risk: OneFan is a web application that handles sensitive user data and makes transactions possible, therefore it could be a target for attackers. Some possible threats are SQL injection or cross-site scripting (XSS) attacks that damage the database, unauthorized access to admin features, stealing user credentials or session tokens, or an attacker trying to change the listing or approval process (for example, by adding false authenticity data). Also, because the server has a Solana private key (thanks to R2's mitigation), a breach of the backend might let someone use that key to make fake records. Moderate risk: web apps are often attacked, although employing well-known frameworks and security best practices lowers the risk.
    - Impact: High, because a security breach might break users' trust, let out private information, or even let fake transactions happen.

- Mitigation: The project used secure coding techniques and took advantage of the security features of the frameworks it chose (Spring Security, Angular's cleanliness, etc.). To stop injection attacks, the server checks and cleans all user inputs. For example, it checks that form fields and API requests follow validation standards (such as making sure they are the right length, format, and type) so that harmful scripts or SQL commands are not accepted. The authentication mechanism employs JWT (JSON Web Tokens) for stateless security. This means that the server doesn't keep track of the session state, and each request must include a valid, signed token. This stops attackers from utilizing stolen session IDs. Role-based access control is used on both the front end (hiding admin-only features from non-admins) and, most importantly, the back end (administrative API endpoints are protected by checks that only allow ROLE_ADMIN users, making sure that even if someone made an API call by hand, it would be rejected without a proper token). The principle of least privilege was used at all times. For example, database accounts only have the permissions they need, and the Solana key is stored on the server in an encrypted form. User passwords and other sensitive information are never saved in plain text. Instead, they are hashed with a powerful algorithm (with salt) so that even if the database leaks, the real passwords will not be exposed. The system also keeps track of security-related events, such unsuccessful login attempts or actions taken by an admin, so that any suspect behavior may be found and checked.

4. R4: Privacy and Compliance

- Risk: OneFan has to preserve user privacy and may have to follow data protection laws (like GDPR) because it handles user information including names, emails, shipping addresses for transactions, and so on. There is a chance that the system could accidentally show private information or not entirely fulfill the law. For instance, putting personal information on a public blockchain would be a problem because blockchain data is public and can't be changed. Also, not letting consumers control their data (like deleting it) could be against the law.
- Likelihood: Low during the prototype stage (the user base is small and the technology is designed to limit data exchange), but not zero.
- Medium impact: If privacy issues are not taken care of, users may not trust the service or there may be legal problems in the future.
- Mitigation: The architecture was made such that personal data would stay off-chain and safe. There is no personal information on the blockchain, just a cryptographic hash of the item's authenticity certificate. This is proof of authenticity. The private PostgreSQL database keeps all of the user's personal information safe. It is not written

to the blockchain or made public. The platform only asks for the information it needs. For example, OneFan doesn't ask for a physical address unless it's needed for a buy transaction. Even then, the information is only meant to be shared with the seller of the item, and it might be removed or anonymized later. A simple privacy policy will let consumers know how their information is used. For example, email is used to log in to an account and send alerts, and addresses are used to ship purchased items. None of this information is made public. Users will also be able to delete their accounts, which will remove or hide their personal information from the system (except for data that needs to be kept for legal reasons, which will be separated from personal identifiers). OneFan lowers privacy risks while still using blockchain where it makes sense by only using it for non-personal authenticity records and following conventional web application data practices.

5. R5: Project Timeline and Feature Creep

- Risk: It's a big goal to build a full-featured marketplace and connect it to blockchain in the time frame of an academic or bachelor's project. There is a chance of over-scoping, which is trying to add too many features (feature creep). This could mean that the project is not finished on time or that it does not work as planned.
- Likelihood: Moderate chance (ambitious projects typically want to add "one more feature," especially when new technology comes out).
- Impact: The project's success will be greatly affected if important features aren't finished or improved before the deadline.
- Mitigation: The project stayed on track by clearly outlining the scope in the Statement of Work and keeping to the major requirements. During development, essential functionality was more important than nice-to-have features. For instance, setting up verified listings and a basic buy/sell flow was more important than any elaborate marketplace features. Instead of trying to make the UI seem ideal, the focus was on making it simple and useful. Time was focused on making sure everything was proper and reliable instead of on fancy visual effects. The development process was done in stages, with regular check-ins and milestones to keep track of progress. At one point, the blockchain integration took longer than expected to debug. The team was ready to put off or simplify other parts of the project (such limiting the types of collectibles or skipping a planned user-to-user messaging feature) to stay on schedule. If the architecture was modular, it would be easy to drop non-essential parts without impacting the main flow. Also, some extra time was set aside at the conclusion of the plan for rigorous testing and bug fixing, since integrating parts could cause problems that weren't expected to happen until the project was almost done. This strict control

of the project's scope made sure that the most important parts were finished and working by the deadline.

6. R6: Operational and Scalability Issues:
    - Risk: If OneFan were to succeed and get a lot of customers, there could be operational problems that go beyond the prototype phase. If there are a lot of users or transactions, the system can slow down or some parts might even crash. For example, a lot of new listings or purchasers could put a lot of stress on the server or database, or the Solana network (in production) might be used more than expected. There is also the chance of user disputes or problems in the real world. For example, a seller might not send an item once it is sold, or buyers might question its validity even though there is a record on the blockchain. These things could hurt the platform's reputation or need action.
    - Likelihood:Low during development and demonstration (the prototype is not meant to be used on a large scale).It employs a controlled environment, but these risks become more important if the technology goes into production.
    - Medium impact: Problems with performance or trust in a live environment could make it harder for users to adopt the product or cost money (for example, support).
    - Mitigation: The prototype wasn't meant to be used on a broad scale, but the design did take scalability into account whenever it made sense to do so. The backend employs a strong relational database and can be scaled up or down as needed. Solana was chosen in part because it can handle a lot of transactions (around 65,000 per second), thus the blockchain layer itself is unlikely to be a bottleneck for the foreseeable future. The code doesn't make any architectural choices that would make it hard to scale. For example, it keeps the application stateless (with JWT auth) so that it may be hosted on numerous servers behind a load balancer if the number of users goes up. An admin console is available so that administrators can keep an eye on activities and step in if necessary to help with dispute settlement and other operational problems. All important actions are logged so that they can be checked later. This means that if a problem comes up, like a fake sliding through or a transaction difficulty, there is a record to look into and fix it. The Future Work part of the platform lists ideas like adding an escrow payment system or delivery tracking. These might lower operational risks even more by automating trust mechanisms. By writing out these future demands, the project shows that it is aware of operational hazards and gives itself a place to start dealing with them as the system changes.
7. R7: Data Consistency and Sync Issues :

- Risk: OneFan needs to keep its database and blockchain in sync with each other to keep its data safe. There could be problems if these parts don't work together perfectly. For instance, an item might be tagged "Approved" in the database even if the effort to register its validity on the blockchain failed or is delayed. This would result in a collectible that looks confirmed in the app but doesn't have a blockchain certificate. In the same way, actions that happen at the same time could be dangerous (for example, two customers trying to buy the same thing at the same time because of race conditions, or a buyer buying the item at the same time that an admin is approving it). In certain situations, there could be multiple sales or states that don't match (an item could be recorded as sold for one user but still be available to others).
- Likelihood: Moderate (multi-step procedures usually have some chance of timing problems or partial failures, especially when an external network call is involved).
- Impact: High for trust and data integrity—if authenticity records or ownership are not consistent, it could lower users' confidence and be hard to fix later.
- Mitigation: The design focuses on transactional processes and checks to keep things consistent. When there are several updates to the database, such making an order and marking an item as sold, they are done in ACID transactions. This way, either all changes go through or none do. The system is set up so that the blockchain write must succeed before the item is marked as fully approved in the database. If the blockchain step fails, the listing stays in pending state, making sure that an item can't be shown as "authenticated" without an actual certificate on-chain. The server utilizes the right locks or checks to deal with concurrency. For example, when a buy request comes in, the database verifies the item's availability again using a locking query to make sure that two purchases don't happen at the same time. The system can also prohibit duplication by using unique restrictions, like saying that an item can only belong to one order. The app can also check on-chain data when a page loads, like when it shows an item marked as authenticated. This is possible because the app was built to make sure that the DB and blockchain will eventually be consistent. For example, the backend (or even the frontend through an API call) could get the hash from Solana to make sure it matches the item's ID and data. This method of double-checking (which is optional) makes sure that the state of the database and the state of the blockchain stay the same. In short, putting operations in the right order and exploiting the strengths of relational transactions can help keep OneFan's off-chain and on-chain data from getting split-brain

The OneFan project became far more resilient and reliable by finding and fixing these risks ahead of time. The structured "risk register" was a living document that helped engineers decide what to work on and how to test things every day. For example, security-related items (R3) led straight to an aggressive input-validation campaign, threat-model reviews, and required static-analysis tests for every pull request. Concerns about integration (R1 and R7) also led to a "fail-fast, prototype-early" mindset. Miniature vertical slices of the Solana write-and-read flow were made long before the full user interface was ready. This made sure that the blockchain calls and transaction-confirmation logic were ready by the time the first admin pressed "Approve" in the UI.

This steady, risk-driven pace paid off in real ways. Possible weaknesses, such faulty JWT payloads that may have allowed privilege escalation, were fixed before they even got to staging. Data-consistency guardrails, which were based on the hazards of concurrent access, kept two buyers from ever racing to buy the same single-copy collectible. Even problems with the user experience, like consumers accidentally buying things that were still in Pending Approval, were expected. Defensive UI prompts and explicit server-side checks now make these kinds of blunders very impossible. In short, risks were not seen as things to check after the fact, but as design restrictions that affected the architecture, coding standards, and deployment strategy from the very beginning.

The initiative also created a culture of being mindful of risks that should last beyond this prototype phase. Automated regression tests linked to each risk category keep running in CI pipelines. They let maintainers know right away when a new commit reopens a threat that was previously fixed. Log instrumentation and dashboards were first made to meet R6 operating needs, but now they give real-time telemetry that can show problems that are starting to happen long before end users see them. Finally, the team came up with a simple retrospective ritual: at the end of each sprint, they make a list of new risks, give them a grade, and either reduce them or accept them with a written explanation.

This risk-aware approach turned what may have been a weak proof-of-concept into a strong, production-ready basis. It not only made the first implementation easier, but it also set the stage for long-term growth as OneFan grows to support more users, more complicated payment systems, and more advanced ways to check authenticity in future versions.

6. Requirements Analysis

THE requirements for the OneFan platform were gathered through analysis of of important stakeholders (collectors, sellers, and administrators listed in Section 3) and by looking at what was wrong with the current sports memorabilia market (Section 2). Functional requirements describe specific features or behaviors of the system, while non-functional requirements describe overall quality, restrictions, or performance standards (including security, usability, and scalability). I looked at both types of requirements to make sure that OneFan does what it's supposed to do and gives users a reliable, easy-to-use experience. I describe the requirements below, noting the main actors (Buyer, Seller, Admin) when appropriate. I also show how some of them are shown in the actual user interface.

6.1 Functional Requirements

The functional requirements define the core capabilities of OneFan. These are grouped by major feature areas and user roles for clarity.

User Account Management

- **FR1: Registration:** The system must let new users make an account by giving the required information, such as a username, email address, and password. It needs to stop people from making more than one account using the same email address. When a new user signs up, they are automatically enrolled as a collector (buyer). Depending on the platform's rules, users may be able to register as a seller or ask for seller status during or after registration. However, all users start with at least buyer credentials.
- **FR2: Login:** The platform will check users' credentials (username/email and password) and set up a secure session after they log in successfully. It is important to check that the passwords entered match password hashes that are saved safely. Once the user logs in, they can use protected features based on their role.
- **FR3: Role Management:** The system must include at least three user roles—Buyer, Seller, and Admin—each with its own set of permissions. A single user account can have more than one role. For example, a Seller is also a Buyer. Depending on their role, some actions are limited. For instance, only Admins can authorize new item listings. When a new user signs up, they automatically get the Buyer position. Admins must be able to provide or take away Seller credentials from a user, or users must be able to ask to be promoted to Seller status with approval.
- **FR4: Profile Management:** Users should be able to see and change their profile information, like their shipping address and contact information. They should also be able to change their password. To stop someone from taking over your account, you should have to verify sensitive actions like changing your password or email address

by re-entering the old password or verifying via email. A user's profile will reveal their job and other important information, such as a seller's overall rating or number of sales.

Seller Functionalities

- **FR5: Create Listing:** Users with the Seller role should be able to make new listings for collectible items. When you make a listing, you have to put in a lot of information about the item, like its title, description, category, year/edition, and price (for example, a fixed sale price). To make their claims more believable, sellers can attach pictures of the item and any supporting documentation, like a scan of an existing Certificate of Authenticity or provenance records. When a seller puts up a new listing, the item doesn't get public right away. Instead, it goes into a state of pending approval. Before buyers can see the listing, an Admin must approve it later (see FR8).
- **FR6: Manage Listings:** Sellers should be able to see a dashboard or interface that shows all of their items and their current status (e.g. Pending Approval, Approved/Active, Sold, Rejected). The seller can change or remove a listing that is still pending or live (i.e., hasn't sold yet). If you make certain changes that could affect authenticity, such updating the item description or replacing photographs, an Admin may have to approve the changes again to make sure that authenticity checks are not skipped. You shouldn't be able to change the listing anymore if the item has already been sold or an order is in the works. You may be able to add tracking information, though. If you cancel a listing, it will no longer be available for sale if it hasn't already been sold.
- **FR7: Sales and Orders:** The system must keep track of an order transaction that connects the Buyer and the Seller and the item sold when a buyer buys something. Sellers should be able to see a list of orders for their items that includes all the information they need to fill the order, such as the buyer's delivery address, contact information, the item sold, and the date of purchase. The Seller will be able to change the status of each order. The seller should designate an order as "Shipped" when they send out the physical collectible. They should also give more information, including a tracking number. The platform might also let the buyer and seller talk to each other about the sale in a simple way (for example, letting the seller message the buyer to let them know that an item has been sent), or at least share the contact information they need, as long as it follows the platform's rules. All records of sales should be kept so that the vendor and the people who run the platform can check them when they need to.

Admin Functionalities

- **FR8: Listing Approval (Authenticity Verification):** Admins (platform administrators) will be able to look at each new listing that sellers have submitted that is still in a pending status. The admin may see all the information on the object, including pictures, documents, and descriptions. At this point, the admin's main job is to make sure that the item is real and appropriate before it goes live on the marketplace. This could mean looking at the Certificate of Authenticity (COA) or other documentation that were given, comparing the item to recognized databases or

examples, and maybe even getting in touch with the seller to get more information or verification. The admin will approve the listing if they are sure that the collectible is real and follows all the rules. When you approve a listing, two things happen: (a) the listing's status changes to Approved/Active, which means that purchasers can view it in the marketplace, and (b) the system starts producing a digital authenticity certificate record on the blockchain (see FR11 below). The admin can reject the listing if they detect problems, such the item looks fake or the listing breaks the rules of the site. If the admin rejects something, the system should let them give an explanation or feedback (such "The authenticity documentation is not enough") that is sent back to the seller so they know why it wasn't accepted.

- **FR9: User Management:** Administrators should be able to handle user accounts and roles all around the platform. This includes giving a normal user Seller privileges (or taking them away if they break the rules) and suspending or banning users who do things that are against the rules or are not allowed. Admins should also be able to help users who are locked out of their accounts or reset their passwords. Also, if users report problems (such a buyer saying a seller is bad or saying an item is false after a sale), admins need tools to look into and fix these problems. There must be an audit trail for user management services to keep track of any changes made by administrators, such as which admin banned which user and why.
- **FR10: Monitoring and Maintenance:** Monitoring and Maintenance: The administrators of the platform should be able to use a backend dashboard or set of tools to keep an eye on all system activity and do maintenance work. This includes looking at stats like the number of current listings, the most recent sales, and maybe even system logs of important events. If a problem comes up later (for example, if an item that was sold is later determined to be fake), admins should be allowed to remove or disable listings that have previously been approved. For example, they could need to flag that item and stop future sales of it or similar products. If necessary, the system should let admins step in and stop transactions or orders that are already in progress. For example, if a dispute is resolved in the buyer's favor, the system should let admins cancel an order or issue a refund in collaboration with payment providers. (While refunds in cash might be handled outside the system, the OneFan platform would need to mark the order as canceled or refunded and maybe change any reputation scores or inventories as needed.) For audit purposes, all important admin actions (approvals, rejections, suspensions, removals, etc.) should be logged so that administrative decisions can be held accountable and tracked.

Blockchain and Authenticity

- **FR11: Blockchain Certificate Storage:** For each item that the admin approves as authentic (every listing that is approved for sale), the system will produce a unique digital authenticity certificate and save a link to it on the blockchain. OneFan's design uses a cryptographic hash, like a SHA-256 hash, to show this certificate. The hash contains information about the item's authenticity and verification. This data may comprise the item's listing ID, name, important description points, and the date or time it was approved. It could all be put together into a structured string or file. The platform

will use a custom smart contract on the Solana blockchain (selected for its speed and low cost) to store this hash in a way that can't be changed and can be checked by anyone. There should be a unique identifier (such a transaction ID or a key that links it to the item listing) for each authenticity certificate entry on-chain. The approval process must trigger a blockchain transaction atomically, which means that the item should not be regarded as fully "active" until the blockchain record is properly established. The system shouldn't finish the approval if the blockchain transaction fails for some reason (like network problems). Instead, it should let the admin know that the transaction failed so they can try again or look into it. In short, every approved listing will include an unchangeable proof of authenticity on the blockchain. An item should not be regarded authenticated (or for sale) without this on-chain certificate.

- **FR12: Authenticity Verification (User View):** The platform must make it easy for anyone, especially buyers, to check the legitimacy of an item using the blockchain record. The user interface will show that an item is "OneFan Authenticated" or something similar for each active listing. This means that the item has passed the platform's verification process. The system should show information about the authenticity certificate in the item's detail page, such as a unique certificate ID or a brief hash. It should also let you view the certificate on the blockchain with only one click. You could make this a "Verify on Blockchain" button or link that takes you to the public Solana blockchain explorer for the transaction or account where the certificate hash was kept. The criteria is that even someone who isn't very tech-savvy can trust that the item is real. The platform does the hard work by capturing the certificate and giving the link. If someone wants to, they may also check the blockchain entry themselves. The system could also execute an internal cross-check. When you load an item's page, it could check the blockchain to make sure the stored hash matches the item's data, which would double-check the integrity behind the scenes. Together, FR11 and FR12 make sure that the authenticity of each item can be checked separately and that this checking is seamlessly built into the user experience.

Buying and Transaction Process

- **FR13: Browse & Search Listings:** The platform should let buyers (and unregistered visitors, in a read-only mode) go through all the collectibles that are offered and search for anything they are interested in. A marketplace view should show all approved and active listings, with important information like the item's title, a picture, the price, and maybe even the seller's username and rating. The system should help you filter and search by a number of things, like the name of the item or keywords, the category (like jerseys, cards, balls, posters, etc.), the sport or team, the name of the person or athlete, and other tags that are relevant. This means keeping an index of listings in the database so that search queries may be run quickly. Users who aren't logged in should still be able to search and browse things (to make the marketplace more open). However, they will need to log in to do anything else, including making a purchase or getting in touch with a seller (to safeguard merchants and make sure they are responsible).
- **FR14: View Item Details:** The system must show a detailed view page for each collectible listing. A potential buyer can see everything they need to know about the

item on this page. There are multiple images (for example, showing different angles or close-ups of signatures), a full description, the asking price, the seller's ID and reputation or rating score, and the item's authenticity certificate status (see FR12, which says it is verified by OneFan). The detail view should also show any other important information, like how many are available (though most collectibles are one-of-a-kind, so the quantity will be 1) and shipping information (for example, if the seller ships internationally, what the shipping cost is, or if they offer free shipping, etc., if that information is collected when the item is listed). This detailed view is very important for customers to make smart choices, therefore it needs to be complete and easy to understand. Buyers shouldn't be able to access the seller's personal contact information directly (to safeguard privacy and keep transactions on the platform), although they could be able to ask inquiries using the platform's messaging system if one is available (this falls under FR7 if it is).

- **FR15: Purchase (Buy Now):** The system must let verified purchasers acquire collectibles that are offered in a way that is similar to how e-commerce works. OneFan can work with either a shopping cart model, which lets customers add many goods to their cart and check out all at once, or a direct purchase model, which provides a "Buy Now" button for buying one item at a time. Since many collectibles are one-of-a-kind, a "Buy Now" button is a simple way to do things. When a buyer clicks "Buy Now" on an item's page, the system checks to see if the item is still available (not already sold or being bought by someone else at the same time) and then asks the buyer to confirm the purchase. After the buyer confirms the purchase, the system registers the order (see FR7 for the seller's view) and marks the item as sold, which means that it is no longer available to potential buyers. Using a shopping cart should let the buyer add products to a cart and then go to a checkout page where they can inspect all the items and place the order. In either case, the process of buying should include getting or confirming the buyer's payment and delivery information. Note: An integrated payment gateway might handle actual payments (such charging a credit card or crypto payment), or this academic prototype might just pretend to do so. The system must be able to handle the entire purchasing process, from choosing an item to confirming the transaction, and it must also update any necessary records, such as the inventory/listing status, order history, and so on. The buyer should get an email and an on-screen confirmation that their order was placed successfully.
- **FR16: Buyer Order Tracking & History:** The platform must let buyers see their past and present orders. Once a user in the Buyer role logs in, they should be able to see an order history page or section that lists all the things they have bought, together with important information like the names of the items, the dates they were bought, the names of the sellers, the prices paid, and the current status of each order (for example, Pending Shipment, Shipped, or Completed). The buyer should be able to see the tracking number or link to the carrier's tracking page for orders that include shipping tracking information (given by the seller in FR7). This criterion makes sure that customers may check on their purchases and keep a record of them for their own use. Also, it helps build trust and openness: a buyer can see that an item they ordered has been dispatched and can expect it to arrive. If there are any difficulties (such the item

not coming or not being what it was supposed to be), this record would also be helpful for letting the platform administration know so they can fix them.

To reach OneFan's aims, all of the functional requirements (FR1–FR16) mentioned are necessary. Users can register and use the platform according to their role, sellers can list and sell real collectibles, admins can keep the platform's integrity by checking items and users, and buyers can be sure that they are buying real memorabilia. The next part talks about how the system works and acts, in addition to these main elements.

6.2 Non-Functional Requirements

Non-functional requirements describe the expected qualities and constraints of the OneFan system. These ensure that the platform is not only feature-complete, but also secure, performant, user-friendly, and maintainable in a production-like environment. The key non-functional requirements are outlined below:

- **NFR1 – Security:** The platform needs to apply strong security measures to keep user data and transactions safe. To keep anyone from listening in, all communication between clients (web browsers or mobile apps) and the server must happen via encrypted channels (HTTPS/TLS). Passwords and authentication tokens are examples of sensitive data that must be saved and managed safely. For instance, passwords will be stored as salted cryptographic hashes instead of in plaintext. A typical way to make sure that each call to a protected API endpoint is permitted is to use signed JWT (JSON Web Tokens) or secure session cookies that specify the user's responsibilities and permissions. The authentication mechanism should be stateless and strong. There must be role-based access control in place so that only people with the right responsibilities can do particular things (for example, only Admins may approve listings and only Sellers can make listings). To stop SQL injection, the system needs to check or clean inputs. To stop XSS (cross-site scripting), the web interface and APIs need to encode outputs correctly. To stop CSRF attacks on operations that change the state, the system needs to use secure tokens or headers. All use of the Solana blockchain, including sending transactions using the platform's private key, should be safe. The private key or keyfile should be encrypted and maintained so that only the server application can access it to sign blockchain transactions. OneFan should also set up logging and monitoring for security-related events. For example, failed login attempts, strange admin activities, or attempts to access resources that aren't allowed should all be noted and send notifications to administrators. In short, security is very important to keep the platform's real data, user credentials, and financial transactions safe from bad people.
- **NFR2 – Performance:** The app should be able to handle a reasonable number of users and data without slowing down. Under normal load (for example, with hundreds or even thousands of active listings in the database), typical page interactions (such loading the marketplace listings or an item description) should finish in a few seconds at most. For search and browse functions to work well, the system needs to use rapid querying and indexing. This will make it easy to find what you're looking for. When an admin authorizes a listing, writing a new authenticity record on the blockchain takes a little longer since it needs to be confirmed by the Solana network. People know that

Solana has fast block times (a few hundred milliseconds) and great throughput, so in practice, a transaction can be confirmed in less than a second. The system should be able to use this speed and deal with the delay in a smooth way. For instance, when an admin clicks "Approve" on a listing, the interface can show a short status like "Recording authenticity certificate..." until the blockchain transaction is confirmed and the item is marked as active. Also, if a customer clicks a "Verify on Blockchain" link, the query to the blockchain should come back in almost real time. The Solana blockchain and a strong backend make sure that the extra checks and records don't dramatically lower the quality of the user experience. To make sure that pages still load and activities still happen in a reasonable amount of time without timing out, the platform should be tested under expected peak traffic, as when a lot of people come to the site following a large game or collectible release.

- **NFR3 – Scalability:** OneFan's design needs to be able to grow with the number of users, listings, and transactions over time. The first deployment might be a single-server application with a relational database and blockchain integration (monolithic design). However, the architecture shouldn't have any built-in bottlenecks that make it hard to scale out. This means that if traffic goes up, the app can be put on more than one server behind a load balancer, all of which will share a database. You can scale your application up (by upgrading to a more capable server) or down (by adding additional servers) without making major modifications to the code if you use standard frameworks and a well-structured database schema. Solana, the blockchain part, can already handle a very high number of transactions per second, which is much more than what OneFan would need even if the platform becomes popular (each approved listing creates one blockchain transaction; even thousands of new listings a day would be easy for Solana's network to handle). The design of the application should avoid performance bottlenecks. For example, large image files should be stored efficiently (possibly on a content delivery network or in cloud storage in a real deployment) and not processed in memory unless absolutely necessary. Also, search queries should use database indexes instead of scanning all data. If the number of users rises a lot in the future, the system should let some parts move to more scalable solutions, like shifting the search function to a separate search engine service or employing a distributed cache to lighten the burden on the database. In short, the criteria say that the system must to be able to handle more demand without needing a complete redesign. Adding resources or making small changes to the architecture should be enough to handle expansion.
- **NFR4 – Usability:** The OneFan app needs to be easy to use and understand so that people of all technological levels may utilize it without any problems. The interface should be easy to use and have a clean, organized structure for looking through collectibles. It should also be easy to post an item or buy it. Forms (for signing up, making a list, etc.) should provide instructions and checks to make sure that the seller doesn't forget to enter a required detail about an item. When a seller submits a new listing for approval, the system might show a message like "Your item has been submitted for review." This is an example of how users should get clear feedback on their activities. You will get an email when an Admin checks to make sure it is real. These kinds of notifications let users know that the activity was successful and tell them

what to do next. Anyone who has used an e-commerce site before should be familiar with the buying procedure, which includes browsing, adding items to the basket, making a purchase right away, and checking out. This makes it easier to understand how to use the site. Any addition of the blockchain part must not be intrusive: the platform will use plain language to show that it is real (no technical jargon). Instead of showing users raw hash values or technical terminology, the site may just show a badge that says "✓ Verified by OneFan" and provide them a link to click on for more information. OneFan makes sure that the new blockchain features build trust without making the user experience more difficult by achieving certain usability standards.

- **NFR5 – Accessibility:** The system should be easy for a lot of people to use, even those with disabilities or who use assistive technologies. This involves following well-known web accessibility rules, such the W3C's WCAG standards. In real life, the web app should use font sizes that are easy to read and color schemes that are easy to see for people who are color blind or have trouble seeing. All pictures, especially those of items, should have the right alternative text so that screen readers can tell blind users what they are. It should be possible to utilize the site's navigation with just a keyboard (this is vital for people who can't use a mouse), and screen readers should be able to read the labels on form fields and interactive controls. The style should also be at least somewhat responsive. The main design might be for desktop web browsers, but the material should reflow or have scrollable views on smaller displays so that those with tablets or smartphones can still use the platform. It should work with all major browsers (Chrome, Firefox, Safari, Edge) and OS systems, and any new web features should work well on older browsers. OneFan not only reaches more people by making its product available to everyone and compatible with different devices, but it also follows ethical and legal standards for inclusive design.
- **NFR6 – Reliability:** OneFan must be very reliable in terms of both uptime and data integrity. Uptime means that the system is available to users and doesn't go down unexpectedly too often. This can be done by using strong hosting and managing errors well (such that a small failure doesn't bring down the whole system). More importantly, data integrity ensures that the system's records (listings, orders, authenticity certificates, user details) stay correct and consistent even when things go wrong, as when the system crashes or the network goes down. For any operation that involves more than one update, the application should use transactions. For instance, when an order is placed, the system might need to create an order record, mark an item as sold, and update the records of both the buyer and the seller. All of these changes should either succeed or fail together, so that we don't end up with an order record that doesn't have the item marked as sold or vice versa. Using database operations that follow ACID will assist keep this consistency. If writing a certificate to Solana fails, the system should not mistakenly mark the item as authenticated/approved (as indicated in FR11); it should either try again or tell an admin about the error. The platform should also be able to handle partial failures. For example, if the Solana network is briefly down, OneFan should be able to safely queue or postpone some activities instead of losing data. To be reliable, you need to back up your data often. The database should be backed up on a

regular basis so that you can restore the data in case of a catastrophic failure. For a production system, daily backups or real-time replication would be best. To make sure that there are no race conditions or concurrency flaws that could corrupt data, the system should be tested in a number of different situations, such as when it is being used by numerous users at the same time or when it is doing the same thing over and over. In short, OneFan needs to keep working consistently and protect the accuracy of transactions and information, even when things go wrong.

- **NFR7 – Maintainability & Extensibility:** The OneFan platform is expected to change over time, especially since it came from an academic background and could get better in the future. Because of this, it needs to be easy to maintain and add new features. Maintainability indicates that the codebase and system design should be easy to read, well-organized, and well-documented. This way, developers can easily make changes to the system when they need to. A modular architecture is needed. For instance, there should be a separation of concerns, which might be done with a layered architecture where the display, business logic, and data access are all separate. This modularity would let a developer change the database layer or update the blockchain integration without having to rewrite the whole system. The architecture should also include special features so that additions are possible. For example, all of the functionality related to blockchain might be in one module or service, which would make it easy to replace or upgrade if OneFan later supports a new blockchain in addition to or instead of Solana. Extensibility also means being able to add new features. For example, if the platform wants to support auction-style listings in addition to fixed-price sales in the future, or if a new user role, like a "Verifier" separate from Admin, is added, the system's foundation should make it easy to add these things. The platform should employ standard technologies and frameworks that are easy to understand so that new developers can get up to speed fast. For example, it should use a standard web framework and libraries for activities like accessing databases and keeping the web safe. Configuration files or environment settings should have things like database connection strings or blockchain node URLs so that the system may be easily adjusted or set up in new places without having to update any code. In addition, the code should be well-documented and have comments, especially in complicated parts like the code that handles blockchain transactions. This documentation is very important for any engineer who has to fix or add to that portion of the system in the future. Writing automated tests (unit tests for business logic and integration tests for essential flows) is also a part of maintainability. A strong test suite will find regressions when changes are made, which will make you feel more confident when you update or rework the system. In short, OneFan's standards for maintainability and extensibility make sure that the platform can be easily managed and grown over time without losing code quality or needing to rewrite everything for new features.
- **NFR8 – Compliance and Standards:** To make sure quality and interoperability, the system design and implementation must follow the best practices and standards of the industry. Using standard data formats and protocols is one example. For instance, the OneFan web API (whether it's for external or internal usage) should use JSON for data interchange because JSON is a format that all web services can understand. The

platform should use well-known cryptographic algorithms and protocols to keep things safe. For example, password hashing should follow best practices, such as using a strong hashing algorithm with salting and peppering as needed. The blockchain smart contract should also follow Solana's security and development guidelines to make sure that nothing in the custom smart contract logic puts the network or the funds at risk. The system should be set up in such a way that it is easy to follow the law or rules that apply. For example, if the platform has users in certain areas, it should be able to help them follow data protection rules like GDPR by letting them delete their personal information and keeping all of their personal information safe. From a development point of view, the project should follow coding standards and style guides (for consistency and professionalism, which makes it easier to maintain). Even though end users can't see these internal code standards (such consistent naming conventions and documentation style), they show how well and carefully the system was built. In short, following the standards will make sure that OneFan is constructed on a strong base, which will lower risks and make it work better with other tools and systems. OneFan is meant to be feature-rich, but also strong and trustworthy. It does this by meeting non-functional needs including security, performance, scalability, usability, accessibility, dependability, maintainability, and compliance. These features make sure that the platform will work successfully in the real world and that people will trust it.

By addressing these non-functional requirements – security, performance, scalability, usability, accessibility, reliability, maintainability, and compliance – OneFan is designed to be not only feature-rich but also robust and trustworthy. These qualities ensure the platform will function well in real-world conditions and can gain users' confidence.

In conclusion, the requirements analysis above gives a full picture of what the OneFan marketplace needs to perform and the standards it needs to meet. The functional requirements (FR1–FR16) identify all the important ways that users can interact with the system and its features, such as managing accounts, listing items, making purchases, and verifying transactions on the blockchain. The non-functional requirements (NFR1–NFR8) make ensuring that the system is safe, works well, and can evolve over time. These requirements make it apparent what the design and implementation of OneFan should look like. They also help create a platform where people can exchange sports memorabilia with trust that it is real and that the service is good.

7. Design

THE design of the OneFan platform meets the functional requirements (FR1–FR16) listed in the preceding chapter and gives a clear plan for how to put it into action. This chapter talks about the system's high-level architecture and important design choices, using UML diagrams to show how the system works and how it is put together. It shows use case diagrams to show how each user role interacts and sequence diagrams for important activities. The design aims for a system that is flexible and scalable, and that connects the web application and the blockchain component in a simple way. A layered architecture (with the Angular frontend, Spring Boot backend, database, and blockchain module all in their own layers) makes sure that concerns are clearly separated. This means that components can be changed or added to (for example, updating the database or adding support for a new blockchain) with little effect on other parts of the system.

7.1 User Roles and Use Cases

OneFan has three main user roles: Buyer, Seller, and Admin. Each role has its own set of interactions and rights. The platform's features are set up around these roles and what they want to do. OneFan helps buyers find and buy real sports collectibles. Sellers put collectibles up for sale and keep track of their stock and orders. Administrators are in charge of the platform's operations, checking stuff and keeping the system safe. High-level use case diagrams (Figures 7.1, 7.2, and 7.3) show how each role is supposed to interact with the system, showing what users can do and how they do it. These diagrams show the system's functional requirements and overall interaction logic at a high level, without going into details about the UI or the code that makes it work.

Figure 7.1 – Buyer Use Case Diagram

The Buyer position includes people who look around the marketplace and buy things, making sure that each item is real before they finish the transaction. Here are some important ways that a Buyer can be used:

- **Register Account:** To make a new user account, you need to give certain information, like your username, email address, and password.
- **Login:** Securely authenticate and start a session to access the platform's features.
- **Browse Listings:** Look at all the approved (current) collectible listings on the site.
- **Search Listings:** You can filter or search listings by things like item name, category, player, or team.
- **View Item Details:** You may see more information about a single ad, such as its description, price, seller information (like rating), and pictures.
- **Buy Now :** Start the process of buying something.
- **View Authenticity Certificate:** Look at the certificate of authenticity that is supported by the blockchain and is linked to the item.
- **View Order History:** Look at the list of things you've bought before and keep an eye on the status of your current orders.

These use cases give customers the confidence to acquire things in the marketplace because they know that each transaction is supported by a blockchain-verified certificate of authenticity and a clear record of the item's legality.

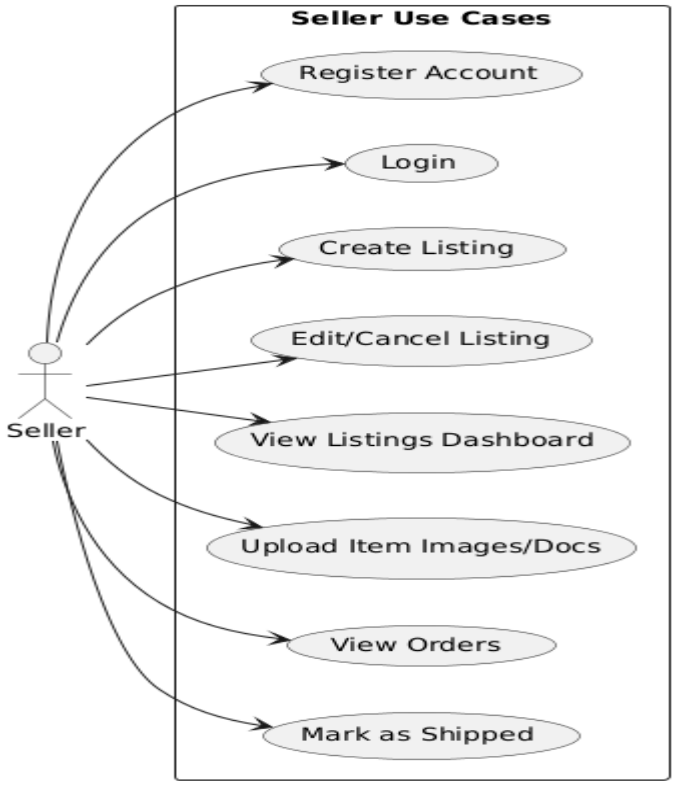

Figure 7.2 – Seller Use Case Diagram

The Seller position includes people who sell collectibles in the marketplace. Sellers are in charge of making listings, keeping them up to date, and shipping orders once a sale is made. A Seller can utilize it for a number of important things, such as:

- **Register Account**: Create a user account, optionally opting to become a seller.
- **Login**: Securely access seller tools and dashboard.
- **Create Listing**: Submit a new collectible for sale by providing item details (title, description, category, price, etc.), along with uploading images and any supporting authenticity documents. New listings are saved as pending approval rather than immediately visible.
- **Edit/Cancel Listing**: Modify details of an existing listing or cancel it (if it is not yet sold). Can change only the price of the collectible.
- **View Listings Dashboard**: View a dashboard of all their listings with status indicators (Pending Approval, Approved/Active, Sold, Rejected.
- **Upload Item Images/Docs**: Attach photographs of the item and documentation (such as scans of physical Certificates of Authenticity or other proof) to support the item's legitimacy.
- **View Orders**: Access information about orders for their sold items, including buyer shipping details and order status, to facilitate delivery.

These use cases give sellers the tools they need to show off and sell collectibles while keeping the platform's standards of authenticity, openness, and customer trust.

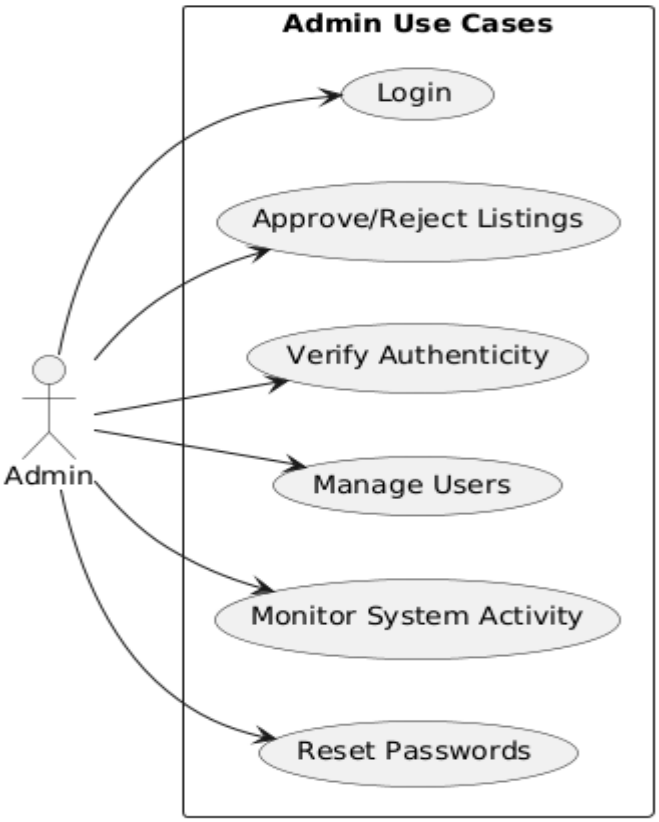

Figure 7.3 – Admin Use Case Diagram

The Admin function includes platform administrators who keep an eye on and fix the system. Admins make sure that listings and users are real and handle the process of checking for authenticity. Some important uses for an Admin are:

- **Login**: Get into the admin dashboard with more permissions.
- **Approve/Reject Listings**: Check out new listings sent in by sellers, along with all the information and papers they need, and then decide if they should be allowed on the marketplace. Approval means the item is real and starts the process of making a blockchain authenticity record (see FR8 and FR11). Rejection means the listing stays off the marketplace, usually with feedback to the seller.
- **Verify Authenticity**: Evaluate item details and support documents to assess legitimacy.
- **Manage Users**: Upgrade buyers to sellers, suspend accounts, reset passwords, or resolve user-reported issues.
- **Monitor System Activity**: Track platform metrics and logs, such as the number of active listings, recent sales transactions, and security audit logs for any unusual activity
- **Administrative Actions:** Perform other maintenance tasks like removing or flagging problematic listings (e.g. if an item is later discovered to be fraudulent) and resolving disputes between buyers and sellers. All critical admin actions should be logged for accountability.

Administrators have full control over platform moderation, quality assurance, and the blockchain integration process through various use cases. This makes sure that all users have a pleasant and safe experience.

7.2 Key Interaction Flows (Sequence Diagrams)

The use case diagrams show what each user role can perform in a static way, whereas the sequence diagrams show how those interactions happen step by step at the system level. This part talks about two important interaction flows in OneFan's design: (1) a Seller making a new listing and (2) a Buyer buying a collectible. A sequence diagram explains how messages are transferred back and forth between the actor (user) and the main parts of the system (the web client, the backend server, external services, and the blockchain) for each flow. These diagrams show the design logic and help you see how the system meets the criteria during important procedures.

Seller Listing Creation Flow

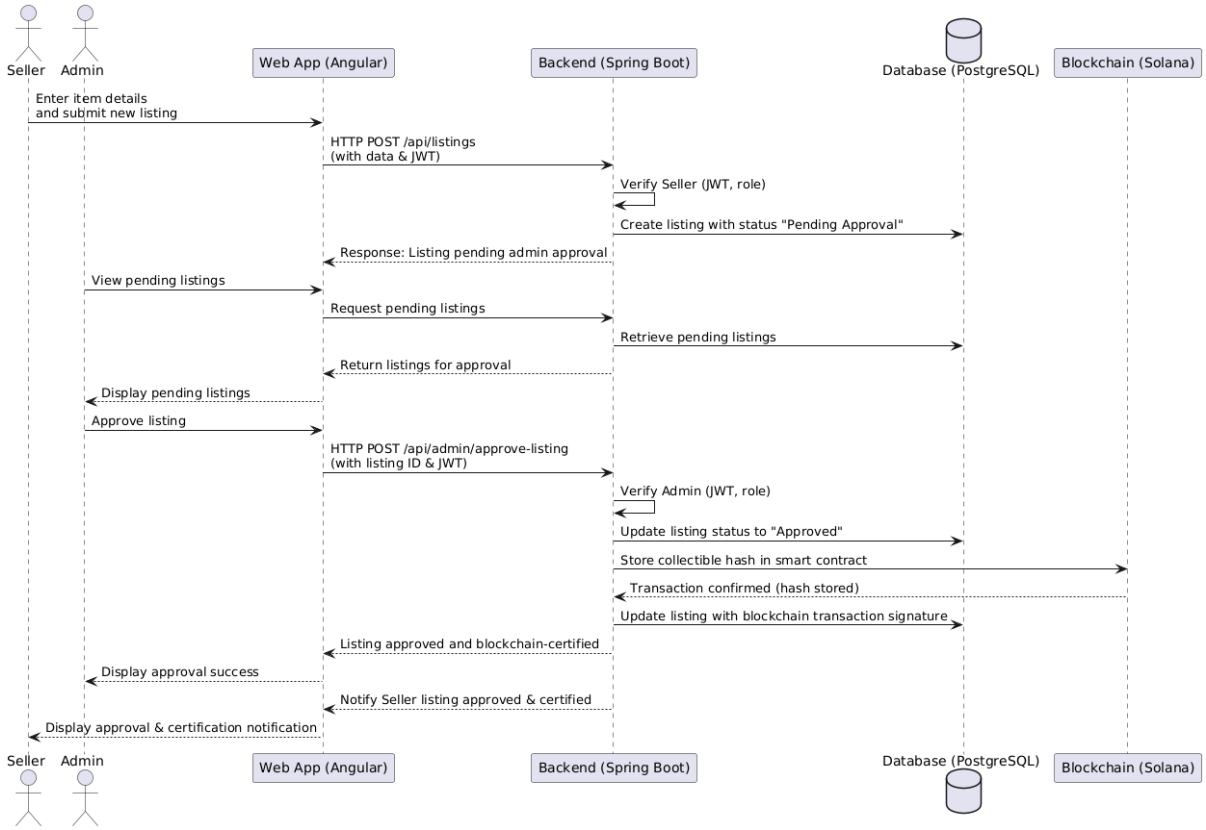

Figure 7.4 – Seller Listing Sequence Diagram

This sequence diagram shows how a Seller makes a new collectible listing on OneFan. The Seller, the Angular frontend app, the Spring Boot backend server, the PostgreSQL database, an Admin (for permission), and the Solana blockchain network (for storing the authenticity certificate) all take part in the interaction. This is how the workflow goes:

1. **Initiate Listing:** The Seller fills out the new listing form on the web frontend with the item's title, description, price, photos, and other information.
2. **Client Request:** The Angular client (which is logged in as the Seller) sends an HTTP POST request to the backend .This request includes the listing data and the Seller's JWT for authentication.
3. **Server Processing (Pending):** The Spring Boot backend checks the Seller's JWT and permissions. Then it saves the listing information in the PostgreSQL database with a "Pending Approval" status. The item is saved in the database at this point, but purchasers can't see it yet and it hasn't been recorded on the blockchain yet.
4. **Admin Review Required:** An Admin user uses the admin interface to get a list of pending listings that need to be approved. The Admin checks the item's data and uploaded documents to make sure they are real and follow the rules of the platform..
5. **Admin Decision (Approval):** The Admin decides to approve the listing (alternatively, they could reject it). If approved, the Admin’s interface triggers an API call to the backend. The server verifies the Admin’s credentials and then updates the listing’s status in the database from “Pending Approval” to “Approved.”

6. **Blockchain Certification:** Once approved, the backend creates a cryptographic hash that shows the item's authenticity data (such a hash of the item's ID and key details or a certificate file) and then calls the OneFan smart contract on the Solana blockchain to save this hash. This blockchain transaction makes a certificate of authenticity that can't be changed and lasts forever. A unique transaction signature or ID is sent back once the transaction is confirmed on Solana.
7. **Update Listing with Certificate:** The backend updates the database's listing record and adds the blockchain transaction reference or signature to it. This connects the listing to proof of its authenticity on the blockchain.
8. **Admin Notification:** The backend sends a confirmation response to the Admin's interface to let them know that the listing was accepted and that the authenticity certificate has been recorded on-chain. The listing is now shown as approved on the Admin interface.
9. **Seller Notification:** You may additionally update the Seller's interface (right away with a push message or the next time they check in) to let them know that their listing is now live and approved. The collectible is presently for sale to the public and has a "Verified Authenticity" label.

The new listing is now completely active on the site, and it has a certificate of validity that can't be changed because it is supported by the blockchain. Buyers can find the item and examine the on-chain record to make sure it is real. This design makes sure that every approved item in OneFan has a way to confirm that it is real, which increases user trust in the marketplace.

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Buyer Purchase Flow

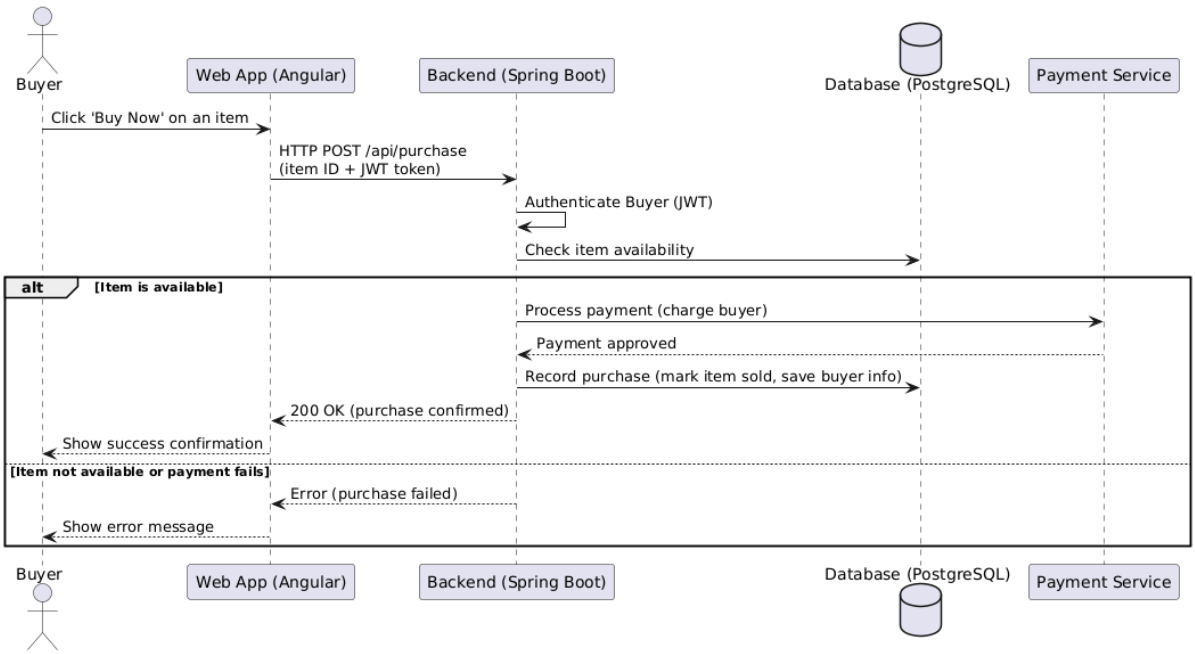

Figure 7.5 – Buyer Purchase Sequence Diagram

This sequence diagram demonstrates how a buyer buys a collectible. It includes the Buyer, the web frontend, the backend server, the database, and an outside payment service (because processing payments is part of making a buy). The primary flow goes like this:

1. **Initiate Purchase:** The Buyer chooses to buy something, like by clicking on a listing's "Buy Now" button. The front end will usually show a confirmation step, like a summary of the order or a request to confirm the purchase.
2. **Client Request:** When the buyer confirms, the Angular frontend makes an authenticated HTTP POST request to the backend, such as to the /api/buyer endpoint. This request includes the item ID and the Buyer's JWT token. This request basically says, "Buyer X wants to buy Item Y."
3. **Server Verification:** The Spring Boot backend checks the Buyer's JWT first to make sure they are who they say they are and what roles they have. After that, it makes sure the item is still for sale by checking that the listing's status is "Approved/Active" and that it hasn't previously been sold or added to another user's basket. At this moment, the server may lock or reserve the item so that other people can't buy it at the same time.
4. **Payment Processing:** If the item is in stock, the backend talks to a payment gateway or service to process the payment. (In the prototype, this may be a fake payment service or a placeholder integration.) The server sends the buyer's payment information and the amount of the purchase to the external Payment Service, which then handles the transaction (for example, by charging the buyer's credit card or crypto wallet). The payment service subsequently sends back a message saying whether the payment was successful or not.
5. **Update Records:** When the payment goes through, the backend updates the OneFan database. The listing is marked as "Sold," and the Buyer's user ID is saved as the new owner of the item. An order record is made that includes information like the date of purchase, the final price, the IDs of the buyer and seller, and so on. The authenticity certificate on Solana doesn't need to change, which is important because it was made at the time of listing approval to prove that it was real. OneFan's approach employs a hash on Solana instead of a transferable token. Ownership transfer is handled in the off-chain database. The blockchain entry is still a permanent confirmation of authenticity and origin, but the platform records now show that the object is now owned by someone else.
6. **Response to Client:** The backend sends a message to the frontend telling it what happened. A confirmation, like an HTTP 200 OK containing an order confirmation or receipt data, is sent back to show that the purchase was successful. If something goes wrong, like the money not going through or the item not being accessible, an error response is returned instead. For example, an HTTP error number and message that explains the problem.
7. **Client Feedback:** The frontend takes care of the backend's answer. If it works, it might take the user to a confirmation page or the buyer's order history, where they can see specifics of the purchase that was made. Most of the time, the UI will show a message that says "Purchase successful," and it may also give the buyer an order number or a

link to see the item's authenticity certificate (which lets them check the on-chain record of authenticity).

- **Alternate – Failure Handling:** If the purchase fails (for example, because there isn't enough money or because someone else bought the item a few seconds earlier), the frontend will show an error message to the user, such as "Purchase could not be completed: item no longer available" or "Payment failed, please try again." If the failure was due to payment problems, the item will still be accessible for others to buy. If another buyer was able to buy it first, it will be presented as sold, making sure that the UI appropriately reflects the current condition.

This purchasing flow is meant to make sure that transactions are safe and trustworthy. JWT authentication makes sure that only a logged-in, authorized user may start a purchase, and the connection to a payment provider (even if it's just a mockup in the prototype) keeps financial information safe. All results, whether they are good or bad, are handled smoothly, giving the user feedback and keeping the system's state consistent. Now that the system architecture and design are set, the implementation may be done with confidence. Chapter 8 will go into depth about how this design was put into action in the working prototype. It will cover things like the user interface and how the backend and blockchain parts were put together to meet the goals of the OneFan platform.

8. Implementation

The implementation phase of the OneFan project turns the design into a working web app that meets all the requirements. This chapter talks about what the platform can do, including screenshots of the running application to show what it can do. The focus is on how the system works from a user's point of view, such as how major features look and work, rather than on the programming that makes it work. Each of the next subsections focuses on one of the main parts of the application (authentication, administrator console, seller tools, buyer interface) and shows what the project looks like now that it is finished. All of the pictures in this chapter are screenshots of the live program that exhibit the user interface and feedback in typical situations. We start this section with the Authentication System, which includes signing up and logging in. This is where all users start, and it makes sure that only people who are allowed to use the platform's features may do so. The implementation focuses on making validation and role management easy for users during sign-up, as well as giving clear feedback for mistakes made during registration and login.

8.1 System for Authentication (Signing Up and Logging In)

Users must sign up for an account and connect in to the OneFan platform before they can fully access the marketplace. The authentication module includes the registration process, in which new users enter their information and choose a role, and the login process, in which current users enter their credentials to access the system. As stated below, these processes have been put into place with a lot of thought given to input validation, security, and ease of use.

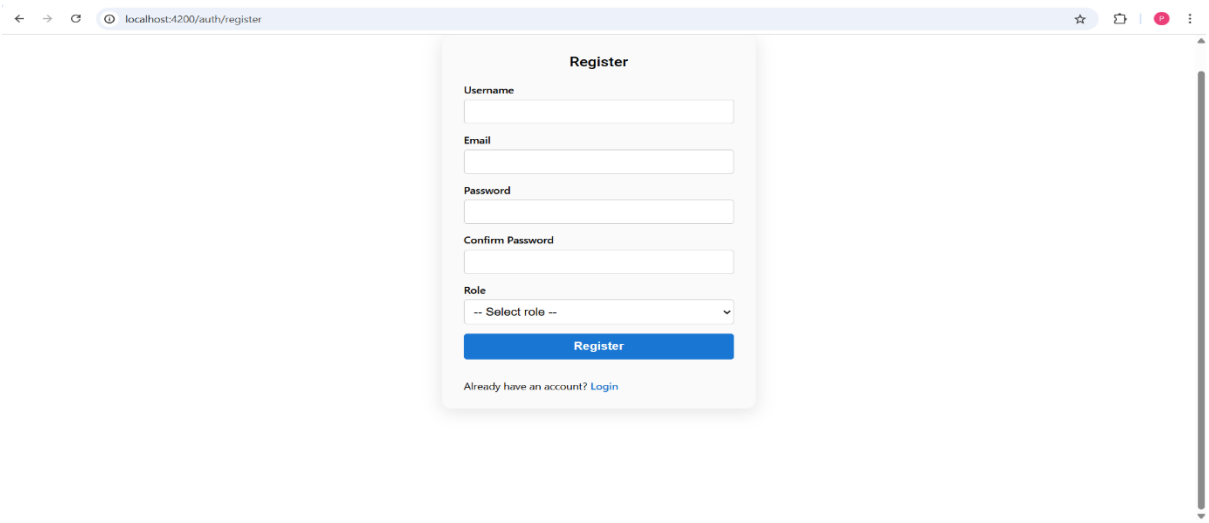

Figure 8.1 – Registration form

Figure 8.1 shows the registration page (in its empty state) where a new user can sign up for an account. The form asks for a username, an email address, a password (with a field to confirm that it is typed correctly), and a role. The Role choice enables the user choose what kind of account they want. By default, a user is a buyer (collector), but if they want to sell things, they can choose to register as a seller. In this interface, "Admin" is also an option, but in a genuine deployment, users wouldn't be able to choose administrator accounts. This way of choosing a role makes sure that when someone signs up, the system knows what role they want to have and can give them the right permissions. There are placeholders and labels on the form to help you fill it out, and the Register button doesn't work until all the needed fields are filled out correctly. There is a link at the bottom that takes current users to the Login page.

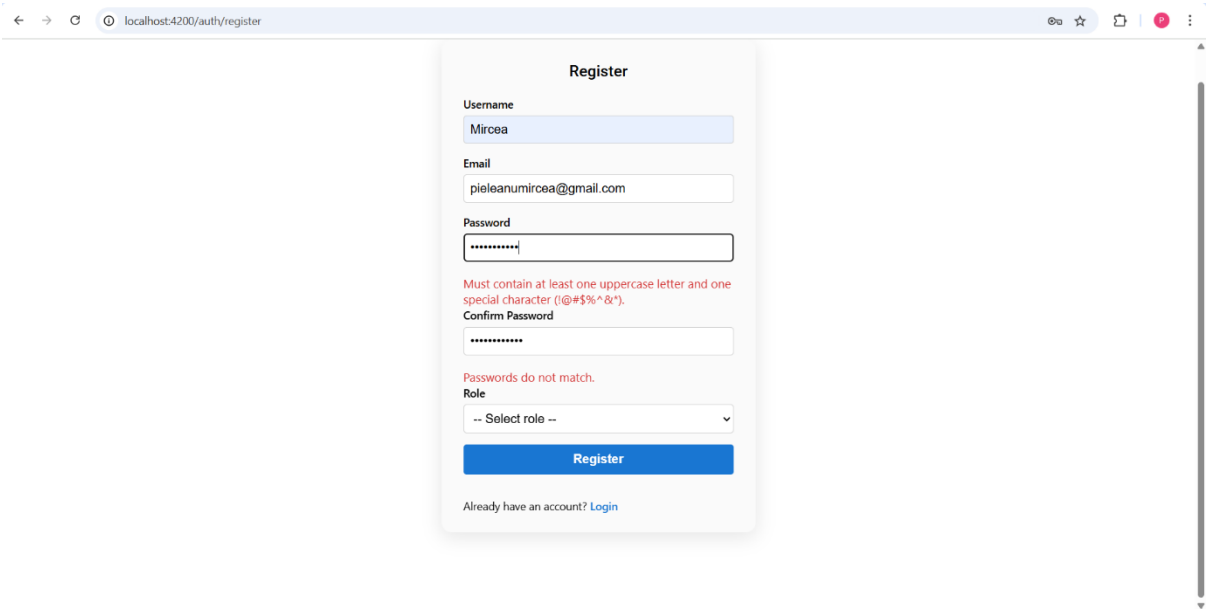

Figure 8.2 – Registration form highlighting password-strength and confirmation issues

In the figure 8.2 if the input criteria are not met, the registration form shows validation errors in real time.In this scenario, the user's chosen password doesn't meet the complexity criterion (which says it needs to have at least one capital letter and one special character), and the confirmation password doesn't match. Under each field, both problems are immediately displayed in red, which gives the user clear feedback on what has to be fixed. Until all of these validation conditions are met, the system won't let you create an account.

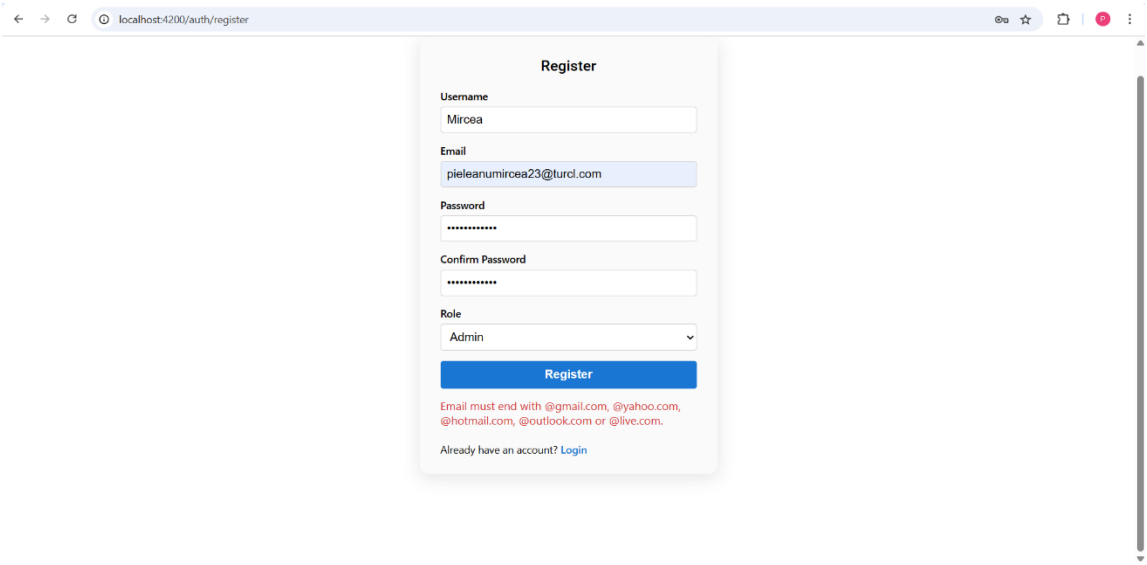

Figure 8.3 – Registration form warning for unapproved email domain

Figure 8.3 shows an example of how to check the validity of an email domain during registration. The platform says that the email must be from a permitted domain.The user's email (pieleanumircea23@turcl.com) is not from one of the approved domains, thus an error message in red says, "Email must end with @gmail.com, @yahoo.com, @hotmail.com, @outlook.com, or @live.com." This limit helps keep user emails trustworthy and illustrates that the system can impose specific validation criteria.

To finish the registration process, the user must give an email address with a valid domain. OneFan makes sure that only real users can get on the platform and that they do so with credentials that have been properly checked by adding these authentication measures. Not only does the registration procedure generate a safe account, but it also gives the user the right role (buyer, seller, or admin), which controls what features they may use after they log in. After users log in, they are taken to the next steps of the program (their dashboard or related interface). From there, they can use the platform's features based on their role.

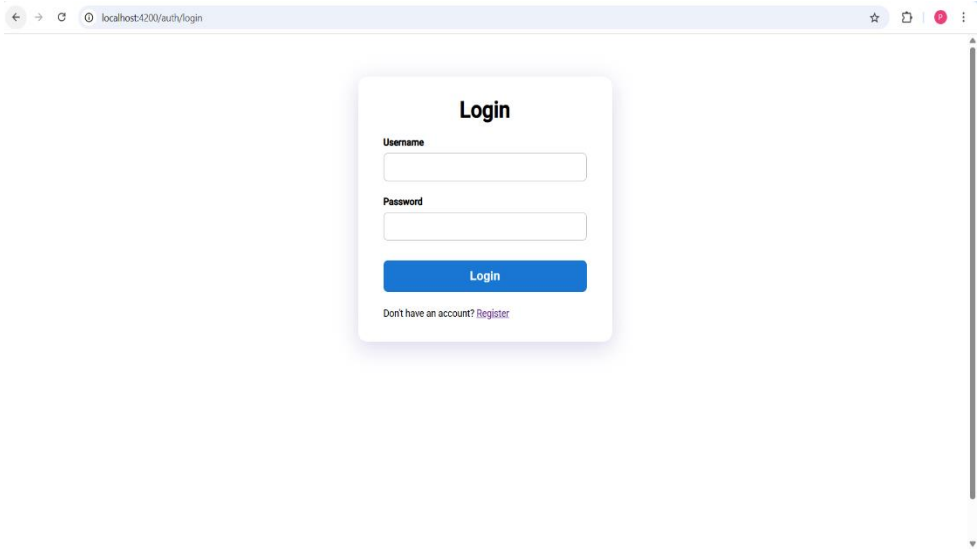

Figure 8.4 – Standard Login Screen

Figure 8.4 shows the normal login screen for OneFan. The interface is straightforward and to the point: it asks for the user's Username and Password, and then there is a blue Login button. There is a link to Register for new users who don't have an account yet. When the user enters proper credentials and clicks Login, they are verified and allowed to use the platform. A session token is sent out behind the scenes, but from the user's point of view, the effect is that they go to their dashboard or homepage. The design respects security best practices by only showing the generic error in Figure 8.5 when someone tries to log in but fails

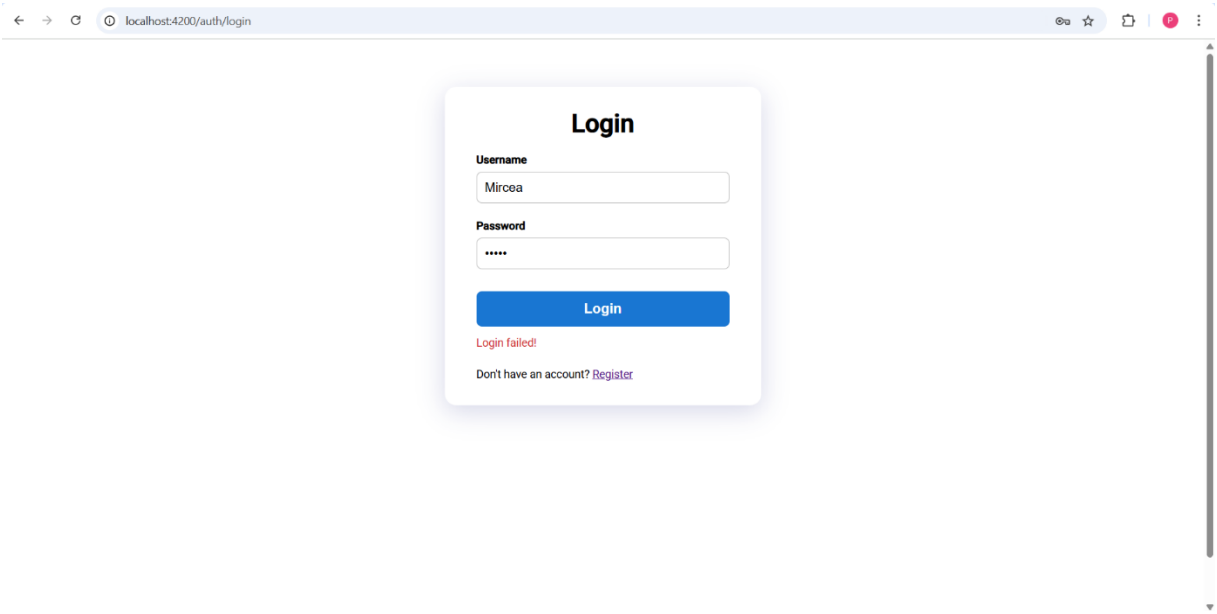

Figure 8.5 – Failed Login

8.2 Admin Interface

OneFan gives admins a special dashboard that shows all of the moderation tools in one place. This helps keep the platform's integrity. From here, the admin can look over new collectible submissions, check their validity or cancel it, manage all of the users, and keep an eye on an unchangeable activity log that is created every time something is approved or denied. The screenshots below show the key parts of this dashboard and how they help with running the marketplace on a daily basis.

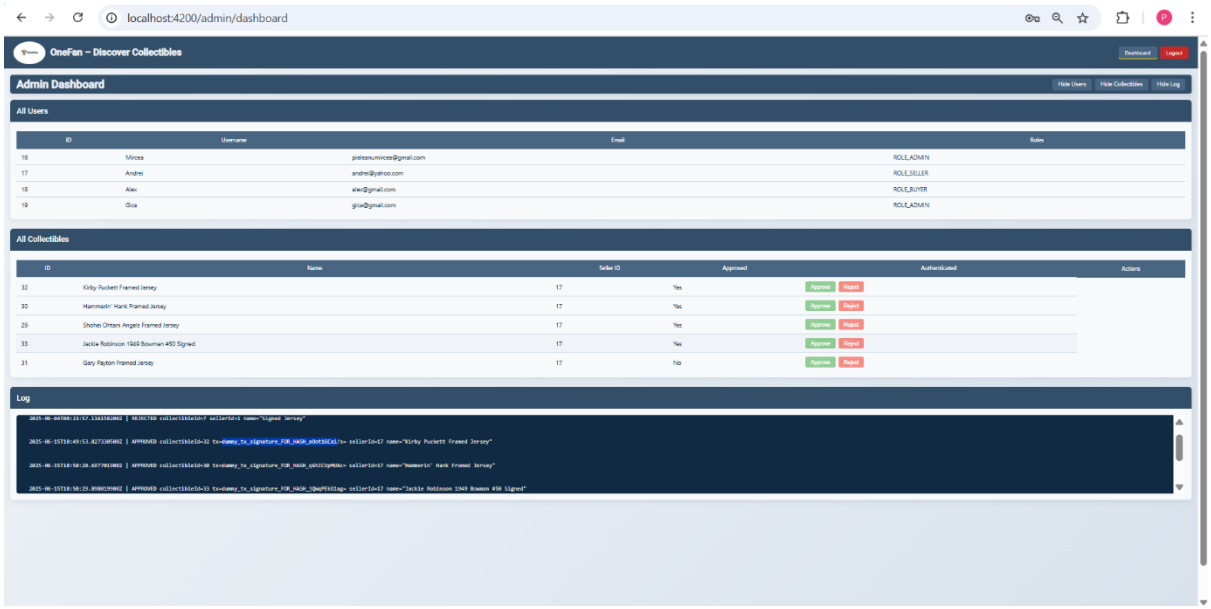

Figure 8.6 – Admin Dashboard

The OneFan admin dashboard is a central place for platform administrators to manage users and collections. When you log in as an Admin, the dashboard shows an All Users area at the top that lists all of the registered users, their email addresses, and their roles (for example, buyer, seller, or admin). This quick look lets the administrator keep an eye on the users and their rights on the platform. The interface is comparable to the main site, so it looks and feels the same, but it has extra controls that only Admin accounts can see. There are three toggle buttons at the top right of the dashboard panel that say "Hide Users," "Hide Collectibles," and "Hide Log." The admin can use these buttons to reveal or hide each component of the dashboard to help them focus and manage space better. For instance, if the administrator wants to focus on going over new collectible submissions, they can conceal the user list and log panels, and the other way around. This style is versatile, so the Admin may simply change the display to focus on the task at hand, even when the amount of data expands (more users or longer logs). The Logout button is also easy to find, so the Admin may safely end the session when they're done with their administrative chores. This is vital for keeping the privileged account safe.

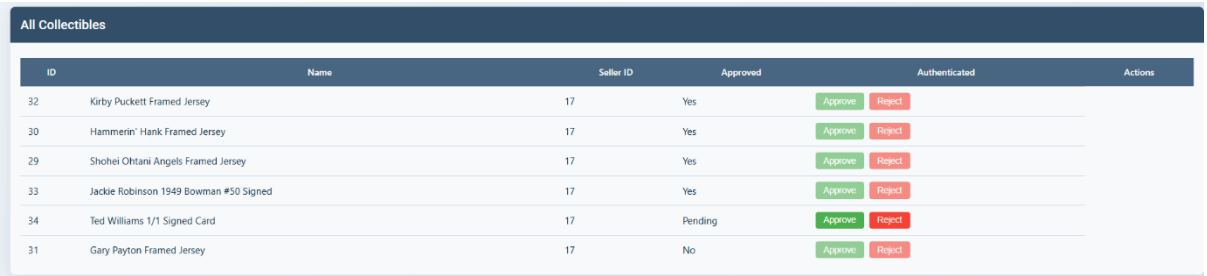

Figure 8.7 – Collectibles Management Table

The collectibles part of the admin dashboard shows all the collectible items that have been sent to the platform and shows whether or not they have been approved. In this table, each

entry shows important information, such as a unique ID, the Name of the collectible (for example, Kirby Puckett Framed Jersey), the Seller ID of the user who submitted it, and an Approved status column. When a seller first submits a collectible, it is marked as "Pending," which means it is waiting for an administrative review. The admin can then click the green or red action buttons on the right to either approve or reject the item.

There are several collectibles listed in the example, along with their current status. Some have already been authorized (marked as "Yes" in the authorized column), one is currently waiting for review, and one has been rejected (marked as "No"). When you approve an item, the system puts it on the blockchain, which proves that it is real. When the item is authorized, the interface changes its status to "Yes." It may also show if the item has been blockchain-certified (in this case, there is a "Authenticated" column to show whether the on-chain record was generated). If you reject an item, it will be marked as not approved, which means it won't show up in the public marketplace. The admin UI manages this clearance workflow, which makes sure that only valid, vetted collectibles are available to purchasers. This directly supports the platform's purpose of trust and authenticity in the marketplace.

Figure 8.8 – Log Panel

The Log panel on the admin dashboard keeps track of all important admin activities and system occurrences, acting as an audit trail. There is a timestamp on each log entry, and it briefly describes the incident. For example, when an admin approves a collectible, a log entry is made that shows the collectible's ID, the blockchain transaction signature that goes with it, the seller's ID, and the name of the item. A transaction signature a collectible that was approved is an example entry in the log. This signature is maintained on the Solana blockchain as proof of validity. In the same way, an action to reject would add an entry with the item ID and name, like the one at the top of the log in Figure 8.9. This logging in real time makes things clear and accountable: administrators can check to see if their actions were logged, and it makes a record of how each collectible was handled that can be followed. The log, along with the user and collectibles sections, enables the Admin to keep an eye on the platform's operations at a high level, from user registrations to listing approvals. This makes OneFan's marketplace implementation more trustworthy and honest.

8.3 Seller Interface

The seller-facing part of the OneFan platform gives customers a dashboard where they can offer memorabilia for sale and keep track of such listings. Sellers can make new collectible listings by filling out a detailed form. They can then check the progress of each submission and make changes or deletions as needed. The system meets all of the seller's needs by providing

an easy-to-use online interface that allows them to create listings, change pricing, delete products that haven't sold, and see the status of their listings. A logged-in seller's navigation bar has a Dashboard link (to rapidly get to the page where they may manage their listings) and a Logout button (to safely end the session). This makes it easy for sellers to switch contexts and exit the platform when they choose.

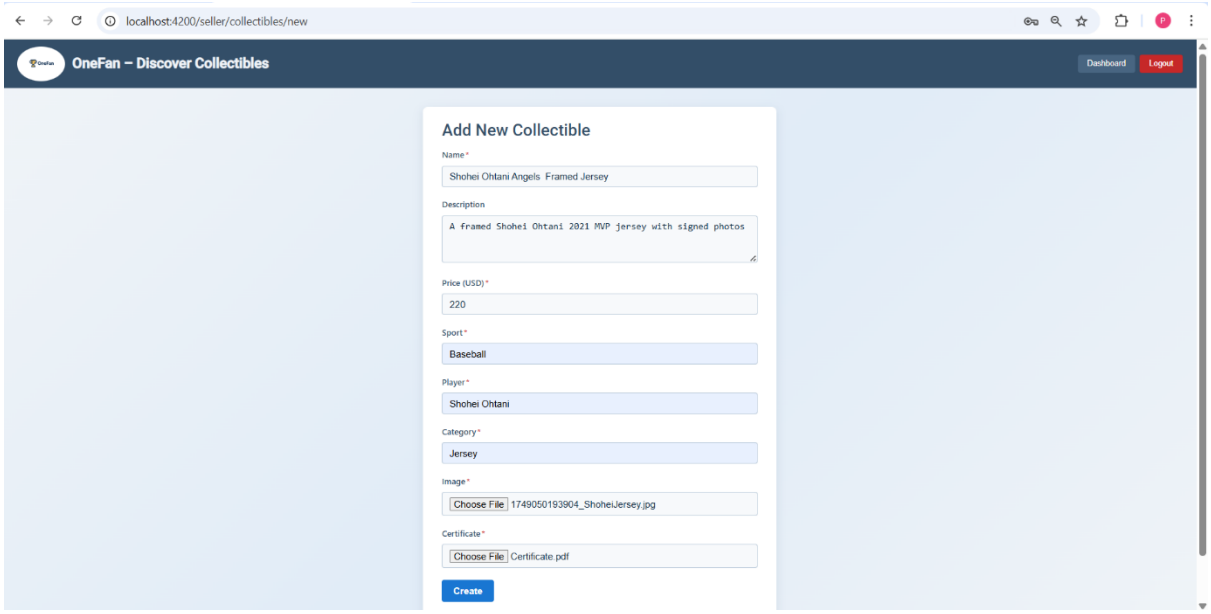

Figure 8.9 – Add new Collectible

The seller must enter all important information about the item in order to create a new collectible. You must fill out all the fields highlighted with an asterisk. These include the collectible's Name, a descriptive Description, the Price (in USD), the Sport (like Baseball or Football), the Player linked with the item, and a Category (like Jersey, Card, etc.). The seller must also attach a picture of the item and a file (in PDF format) that proves the item is real. These accompanying files, like a scanned paperwork and a picture of the item, are very important since they give the admin proof and documentation to look over. The form makes sure that all the necessary information is gathered up front to help with the authentication procedure. The seller submits the new listing to the system after filling out the form and clicking the Create button. At this point, the listing does not become immediately public. Instead, it enters a pending approval state , meaning it awaits an administrator's review for authenticity compliance. This design guarantees that every item undergoes an authenticity check by the admin before appearing to buyers on the marketplace, aligning with the platform's trust and safety requirements.

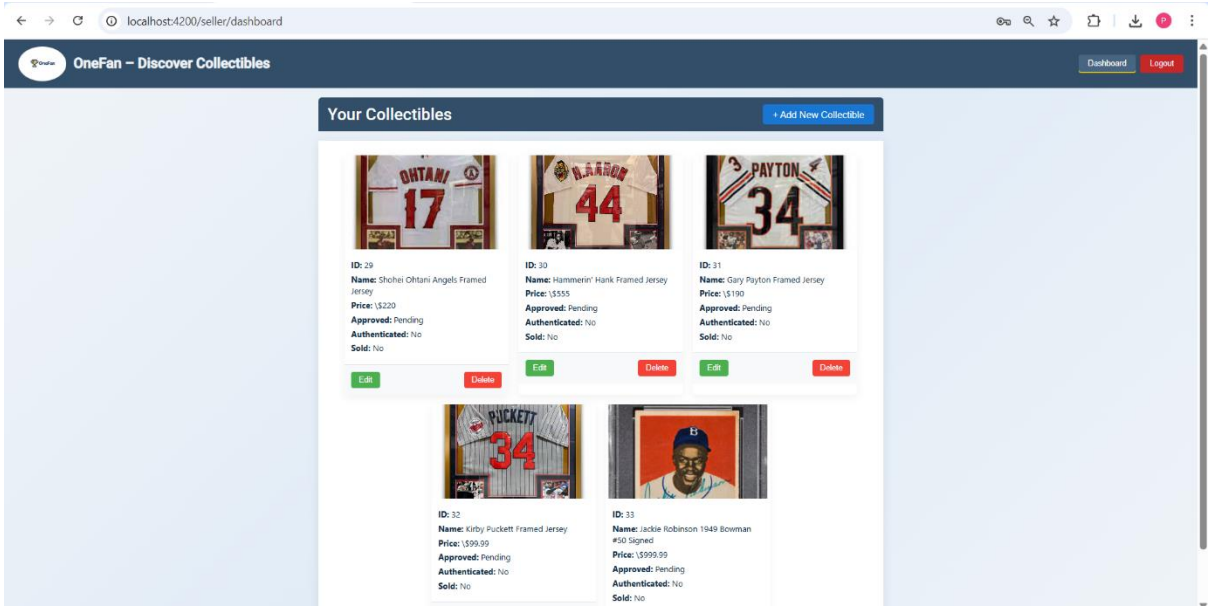

Figure 8.10 – Seller Dashboard

Sellers may keep an eye on all of their listings using the Seller Dashboard when they submit them. The dashboard is a special view that shows all the collectibles the seller has submitted. Each one is shown as a card with important information and status indicators. The dashboard shows the unique ID, name, price, and three crucial status attributes for each item on the list: Approved, Authenticated, and Sold. The "Approved" status shows if the admin has looked at the listing. New items will show "Pending" (waiting for permission) in this field, and after an admin approves the listing for sale, it will change to "Approved".

The Authenticated status shows if the item has been given a blockchain-backed authenticity certificate. This would usually change to "Yes" after an admin approves it and the Solana blockchain hash is recorded for the certificate.

The sold status only reveals whether or not a buyer has bought the item. It says "No" for available listings and "Yes" for sold items.

This status display, which shows everything at a glance, meets the necessity for sellers to keep track of all their listings in one spot, whether they are pending, approved/active, sold, or rejected. The Dashboard button on the top navigation bar lets you see the dashboard view at any moment. This shows that it is the main place for seller activity.

Figure 8.10 shows the Seller's Dashboard with Listings: The seller's dashboard shows each collectible listing as a card with information about it and its current status. This example shows several submissions, each with a card that lists the item's name (like "Shohei Ohtani Angels Framed Jersey"), the set price (like "$220"), and the status flags for Approved, Authenticated, and Sold. The new items in the image are all listed as "Pending" under "Approved" and "No" under "Authenticated." This means that they are waiting for admin approval and have not yet been certified on the blockchain. Everyone has sold "No" since none of these things have been bought yet.

This interface lets the seller keep track of how each listing is doing. For example, the seller may see if an item is still waiting for review or has been authorized and is now visible to buyers. If an admin turned down a listing, the Approved status would disclose that (for example, it

would say "Rejected"). However, the platform does not offer sellers clear messages about approval decisions. Instead, it's the seller's job to keep an eye on these status changes on their dashboard.

The dashboard also has action buttons on each item's card, such as Edit and Delete, that let the seller easily manage their listings. The dashboard page also has a "Add New Collectible" option at the top.

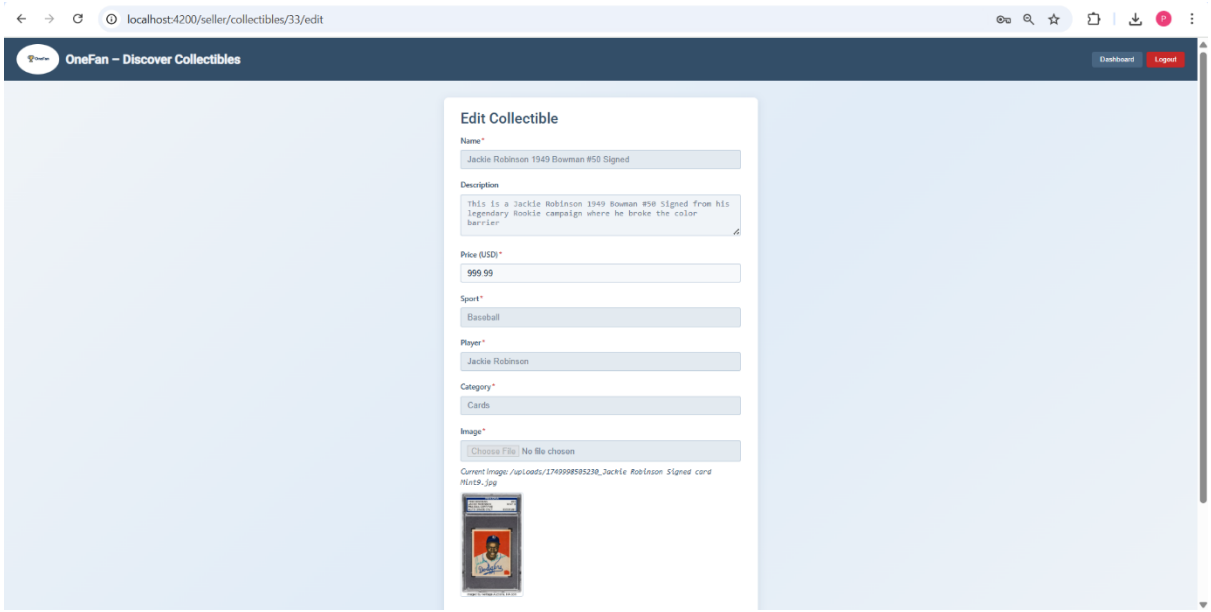

Figure 8.11 – Edit Collectible

This lets the seller rapidly make a new listing if they have other goods to sell. Sellers can change a listing after it has been made, however in the OneFan implementation, this power is limited on purpose to stop abuse and keep faith in the authenticity check.

In particular, after a collectible is posted, the seller can only change the price field; all other item details are locked and can't be changed. This design choice is based on the need to keep the admin's approval valid. After submission, important fields like the item's name, description, sport, player, category, or uploaded documents cannot be changed on a whim, because doing so could make the original authenticity review less valid. The site only lets users adjust the price, which keeps sellers from making big changes to an item's description or identity after it has been approved without anybody watching. If this were to happen, the item would have to be re-approved. It also matches how things usually work in the market: prices regularly need to be changed (for example, to meet demand or fix a mistake in pricing), but changing the essential elements of an item is less often and more difficult.

The seller can change the price of a listing by clicking the Edit button on the item's card in the dashboard. This opens an editing form that is already filled up with the listing's information. The edit form for a collectible has input boxes that are grayed out, which means that all fields except the price are read-only. The seller can change the item's price (in USD), but the fields for Name, Description, Sport, Player, Category, Image, and Certificate stay the same as they were when the item was created. This makes sure that the item's identity and authenticity

documents are always the same as what the admin first approved. The seller can save the change after altering the price.

The new price will show up in the listing (both on the seller's dashboard and for buyers if the listing is already approved and live). It's important to note that this edit doesn't need a new admin permission because only the price can change and the item's authenticity credentials stay the same. If the platform had let users update the description or other fields, on the other hand, it would have had to demand an admin re-check, since large modifications require re-approval.OneFan makes the workflow easier and keeps trust by locking such fields. Buyers and admins may be sure that the seller hasn't changed the descriptive data or proof of authenticity of an approved listing after the fact.

The edit interface also shows the seller how to get about. The Dashboard link at the top (in the figure's header) takes the user back to the overview of their listings, and the Logout button lets the seller leave the session when they are done making changes. Finally, the seller interface lets you delete a listing by clicking the red "Delete" button on each dashboard card (shown in Figure 8.10). If a seller no longer wants to sell an item or made a mistake in the listing and wants to start over, they can delete the listing as long as it hasn't been sold. As part of the seller's management over their inventory, they can delete (or cancel) a listing that is in a pending or active condition. The item is no longer visible on the marketplace and is also gone from the seller's dashboard after it has been deleted. This helps sellers keep track of their collections and make sure that only the goods they want to sell are featured on the site.

To sum up, the seller interface in the OneFan app makes it easy and safe for sports memorabilia owners to add listings. It starts with a form-based entry for a collectible (with all the necessary information and proof of authenticity), then there is a dashboard that shows the status of each listing, and finally, sellers can edit the price and withdraw listings. This implementation strikes a compromise between making it easy for the seller to use and putting in place protections that respect the marketplace's criteria for authenticity and integrity.

8.4 Buyer Interface

OneFan's buyer area makes it easy for collectors to find, assess, and buy authenticated sports memorabilia. Users may go across the whole marketplace from the dashboard, use quick filters to narrow down results by player or sport, see only certified items, and see the status of each listing at a glance.

When someone starts to buy something, safeguard prompts let them know if the item is still waiting for verification. This makes sure that purchasers are fully informed of the risks. After a purchase, every collectible is stored in the My Collection view. This lets customers keep an eye on their collections and see how each item's authentication status changes over time. The numbers that follow show these basic skills.

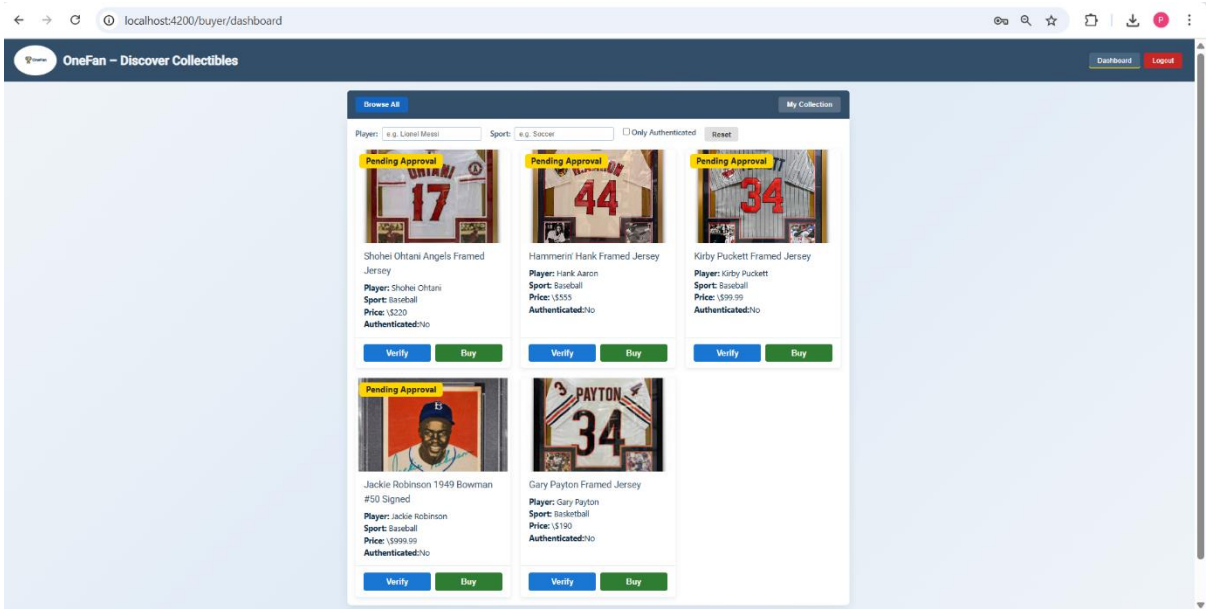

Figure 8.12 – Buyer Dashboard

Figure 8.12 shows the main dashboard for buyers, which shows all the collectibles that are for sale in the marketplace, both authenticated and pending. The listing for each item shows important information including the name, player, sport, price, and current authentication status. There are apparent signs that collectibles are still waiting for verification. For example, unverified products have a yellow "Pending Approval" banner on them (as seen on multiple listings in Figure 8.12), and the item's details say "Authenticated: No." At the top of the page is a navigation bar with a link back to the Dashboard (the main view for surfing) and a Logout button for leaving the account.

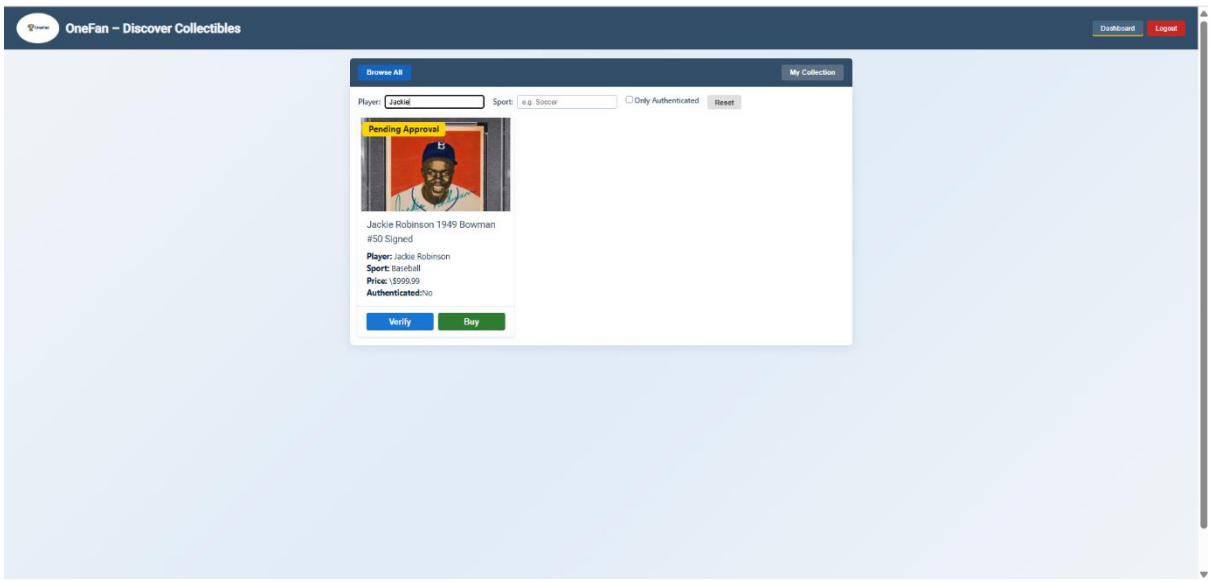

Figure 8.13 – Search Filter by Name

The buyer can search the marketplace by player name to find certain collectibles. You can type a name into a search box called "Player." For example, typing "Jackie" right away narrows the results to items that match. Figure 8.13 shows that typing "Jackie" only brings up

the Jackie Robinson item. This dynamic filtering allows buyers rapidly find memorabilia for specific sportsmen without having to look through all the others collectibles.

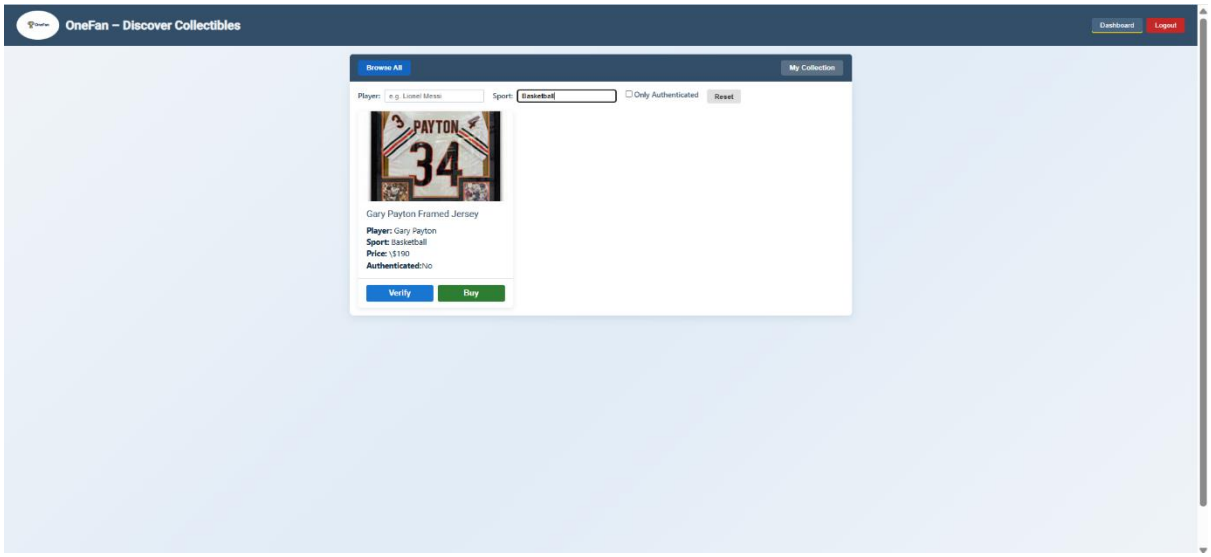

Figure 8.14 – Search Filter by Sport

Buyers can sort collectibles by sport category in addition to searching by name. To see only collectibles from a certain sport, type "Basketball" into the "Sport" filter field. When the list is filtered for basketball items, Figure 8.14 shows that the only result is a Gary Payton jersey. With this feature, customers can search the marketplace for products that are relevant to their sport, such as baseball memorabilia or basketball items only when they need to.

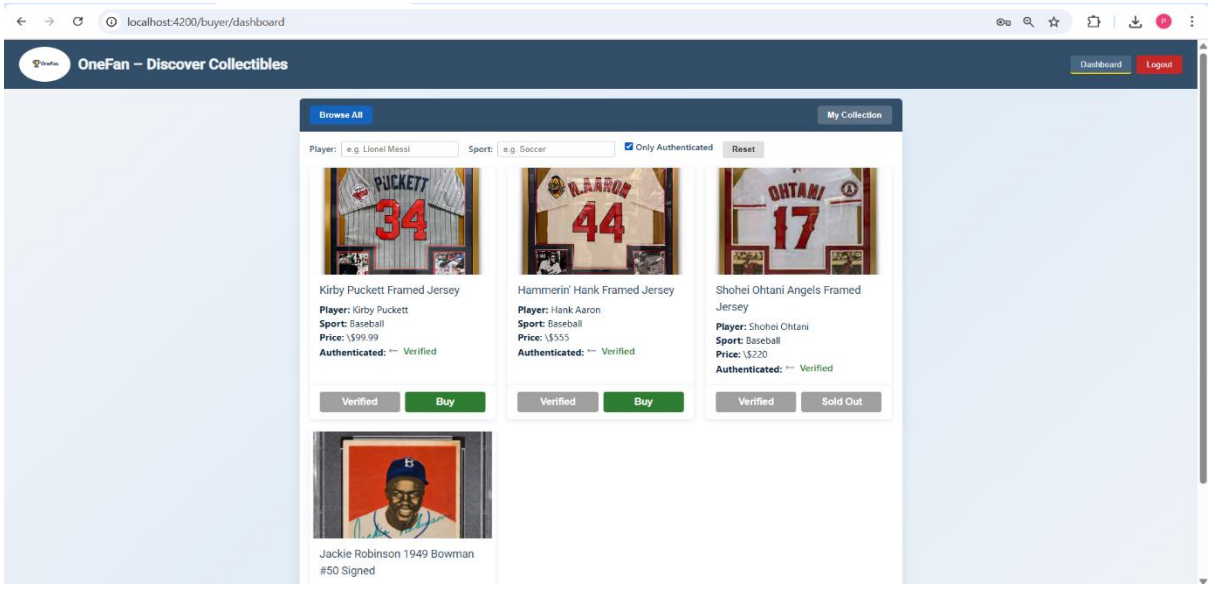

Figure 8.15 – Search Filter for only Authenticated

The buyer interface also lets you show just verified collectibles. When you click the "Only Authenticated" box, the marketplace view only shows products that have been approved by an admin. This hides any listings that are still waiting for approval. In the filtered view that comes up (Figure 8.15), only validated items are still displayed. Each one has a green "Verified" status signal next to its listing. This feature makes it easy for customers who only want to acquire authenticated memorabilia to see only the certified goods.

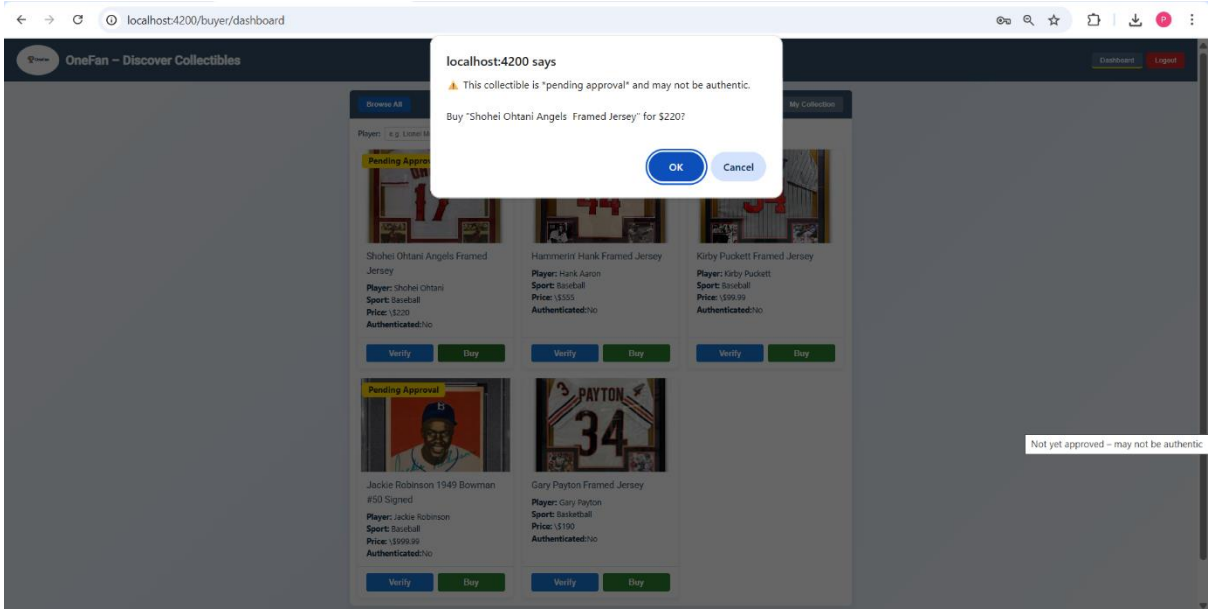

Figure 8.16 – Warning for unauthenticated purchase

Before the sale is finalized, the system shows a warning dialog if a buyer tries to buy an item that hasn't been authenticated yet. This pop-up alert makes it plain to the user that the item is "pending approval" and may not be real. It also asks the customer to clarify whether they want to go through with the purchase by clicking OK or Cancel. This protection makes sure that customers know that the item's authenticity is not definite and that they must explicitly accept that risk before going through with the purchase.

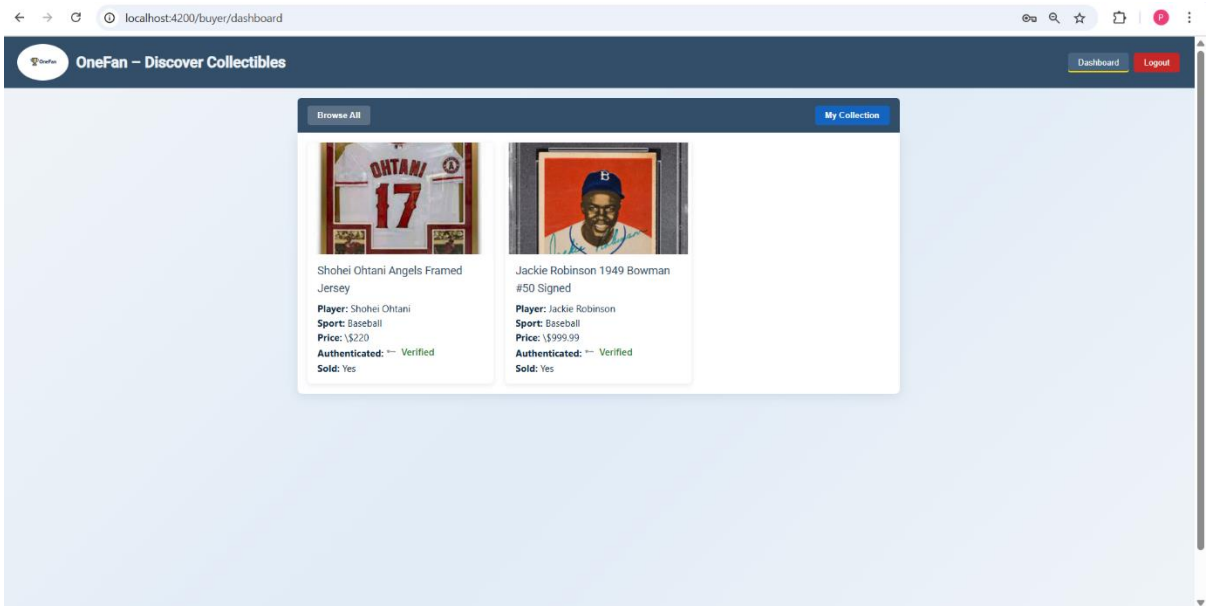

Figure 8.17 – Buyer Collection

Figure 8.17 shows the My Collection page with only purchases that have been properly validated.

When an administrator confirms a listing and the platform saves its authenticity hash on the Solana blockchain, the item goes from the public marketplace to the buyer's private collection

with a green Verified marker. In this case, two legitimate items—a Shohei Ohtani Angels Framed Jersey and a Jackie Robinson 1949 Bowman #50 Signed card—show up as different cards. Each card shows the player's name, sport, and the ultimate price, as well as status tags: Authenticated: Verified means that the blockchain certificate is in place, and Sold: Yes means that the transfer of ownership is complete. The UI around it keeps a familiar navigation bar with "Browse All" for going back to the marketplace, "My Collection" for going back to this view, and "Dashboard" and "Logout" in the header. This makes it easy for purchasers to navigate between contexts. The page works as a digital exhibition and provenance ledger by putting all of the verified purchases in one curated list. This gives collectors quick peace of mind that each piece they buy has gone through OneFan's verification process and is permanently linked to an unchangeable on-chain record.

The implementation phase turned OneFan's concept into a working prototype that shows how all of the platform's important workflows work. The authentication module makes sure that only verified users may do their jobs. Administrators may now accept or deny listings, make blockchain certificates with just one click, see a live activity log, and manage all users from a single dashboard. Sellers get a targeted toolbox that makes sure that all of their listings are full and backed by certificates when they are created, lets them make safe price-only changes, and shows them an at-a-glance view of their inventory for keeping track of approvals, authentications, and sales. Buyers, on the other hand, use an easy-to-use marketplace where they can filter by player or sport, see only verified products, get clear cautions about the risks of pending artifacts, and keep a verified personal collection that also serves as a provenance ledger.

These interfaces show that the platform's main goal—trading sports memorabilia with clear, blockchain-guaranteed authenticity—can be reached without making consumers deal with the complicated nature of distributed ledgers. The prototype not only meets all of the functional and non-functional requirements set out in earlier chapters, but it also lays a strong framework for future improvements like on-chain verification buttons, mobile responsiveness, and advanced payment integration.

9. Future Work

THE existing OneFan platform satisfies its original goals, however there are many ways to make the system better and bigger. The following improvements make sense as the next steps since they expand on what OneFan already does well

Enhanced Security and Verification: Making authentication methods stronger is a top priority for future development. For instance, the platform may connect real-life valuables to NFC chips or QR codes that are linked to the blockchain record and are hard to tamper with. This would make the connection between the real item and its digital certificate even stronger. Also, using multi-factor authentication for user accounts and better managing the keys for any blockchain credentials will help keep people from getting into accounts they shouldn't. These steps would boost users' trust by making sure that only real, verified things are traded on the platform.

Additional User Roles and Features: Adding more ways for users to communicate with one other on OneFan can make the platform more flexible. One suggestion is to add official authenticators or expert curators as a new user position. These are people or organizations who are trusted to check collectibles in person and verify their validity on the blockchain. Some new elements for the marketplace may be an auction system for rare collectibles, a peer rating and reputation system (so that buyers and sellers can get to know each other better over time), and social features like user profiles or community forums for collectors. These kinds of changes would appeal to a wider range of people and keep consumers interested, turning OneFan from a simple marketplace into a lively community for sports collectibles fans.

Integration with External Systems: Connecting OneFan to other platforms and services would make it more useful and make the user experience smoother. For example, connecting with standard e-commerce payment gateways and shipping companies might automate the logistics after a sale is made, from safe payment processing to tracking the delivery of the goods. Working with sports leagues or official merchandise issues could make it possible to verify directly from the source (for example, a football club or memorabilia company could use the blockchain to validate that an item is real). The platform might also let third-party developers access API endpoints, which would let OneFan's verified collectibles be seen or exchanged on bigger marketplaces and even used in augmented-reality fan experiences. These kinds of integrations would help OneFan reach more people and make it fit in better with the rest of the digital commerce ecosystem.

Expanded Blockchain Functionality: Future versions of OneFan might use more advanced aspects of the Solana blockchain to add more functions to the site. Right now, Solana is mostly used to keep track of ownership transfers and authenticity records. In the future, Solana smart contracts could make more complicated transactions easier, like an escrow contract that holds payment until the buyer confirms delivery and authenticity of an item. There is also the possibility of supporting cross-chain interoperability, such as allowing authenticity records to be copied to or moved to other blockchain networks. This might increase the number of users or make the system more resilient. Another option is to look into limited tokenization of collectibles in a controlled way. OneFan intentionally left out standard NFTs in the first design, but future research could look into hybrid approaches (like semi-fungible tokens or "digital twin" certificates) to see if they make items easier to trade while still avoiding the speculation

and fraud problems that purely NFT-based approaches have had. Any such growth would need to keep the main idea of trust by making sure that everything is real.
Scalability and Performance Optimization: For real-world application, it's important that the platform can handle an increasing number of users. Future work should involve thorough performance testing of the web application and backend services while they are under heavy pressure. They should also optimize database queries or caching mechanisms to keep response times fast as the number of users and listings grows. On the blockchain side, as more people use it, the transaction throughput and prices must stay reasonable. This could mean updating to newer Solana features or adding more efficient Solana nodes to process more transactions. Adding monitoring and alerting features to find any performance problems or Solana network problems in real time would also be helpful. Making a web interface or a separate mobile app for OneFan that works well on mobile devices will also make it easier to use and more accessible, which would probably lead to more people using it. OneFan can keep things running smoothly and reliably even if the number of collectibles and trades goes up by taking care of scalability and performance early on.

Each of these improvements would make OneFan's value proposition as a safe, easy-to-use marketplace for sports memorabilia even stronger. The platform can stay relevant and strong in the changing world of blockchain-based commerce by continuing to innovate in technical, functional, and operational ways. This will allow it to keep up with new technologies and user expectations.

10. Conclusion

THIS thesis presented the design and implementation of OneFan, a web application for trading authenticated sports collectibles using blockchain technology. The main purpose was to let buyers and sellers trade sports memorabilia with confidence that each item was real, without using the typical NFT methodology.

I used the Solana blockchain to make a digital record for each collectible that can't be changed, and we gave each item a unique on-chain certificate of authenticity. This helped us reach our goal. The project showed that sports collectibles can be listed, sold, and verified on a decentralized ledger in real time by creating a working prototype. This directly addresses long-standing problems in the collectibles market with fake goods and false claims of provenance.

The user interface, backend server, and blockchain integration all worked together to accomplish the thesis goals. This created a platform that greatly increases trust in exchanging digital sports memorabilia.

The way OneFan built its architecture was very crucial for its method of checking authenticity. The choice of the Solana blockchain was very important since it has a high throughput and low transaction fees. This meant that the system could record and update authenticity information relatively instantly and at very little expense, even as the number of transactions grew. This performance is very different from that of older blockchain platforms, like Ethereum, which had slow and expensive transactions. This makes Solana a good base for a live marketplace where many tiny updates on authenticity happen all the time.

The choice not to show collectibles as transferable NFTs on public marketplaces was just as critical. OneFan avoided many of the problems and risks that come with the NFT ecosystem by not going that path. The platform doesn't treat each collectible as a tradeable token with its own changing market value. Instead, it only uses blockchain to prove authenticity and ownership. This design choice shows that blockchain may be used for more than just "collectibles-as-crypto-assets." It shows a new use case where the ledger's main job is to be an unhackable authenticity register. The end result is a system that gets the benefits of decentralization—security, transparency, and data integrity—without the extra work or problems that often come with NFT-driven projects. The OneFan platform makes people trust the online sports collectibles market a lot more by tackling the problem of authenticity at the transaction level.

By putting each item's history and ownership on a public ledger that can't be changed, it becomes much tougher for bad people to sell fraudulent objects or change transaction histories. Buyers don't have to rely on paper certifications or the reputation of vendors anymore. They can now check the authenticity of an item right away by looking at its blockchain record. Every time a collectible changes hands, the transfer is recorded in a way that can't be changed, making it easy to see who has owned it since it was first listed. This method fits with what people in

the collectibles sector have seen: blockchain-based records (also known as digital certificates of authenticity) help fight fraud and make both buyers and sellers feel more secure.

OneFan essentially offers clear provenance monitoring, which makes the memorabilia industry safer and more open. Participants can trade with the peace of mind that claims of authenticity can be independently verified on the ledger. This greatly lowers the chances of disagreements and uncertainty. This greater openness and safety leads to more confidence among users and a stronger trading environment, which is what blockchain promised to do: "lay the foundation for a more secure, transparent, and accessible market" for valued items.

The results of this study have effects on digital commerce and the use of blockchain that go beyond the immediate focus of trading sports memorabilia. OneFan is an example of how decentralized technology may be used in easy-to-use apps to solve real-world problems. In this case, it was used to check the authenticity of products in peer-to-peer trading. The platform's performance shows that comparable designs could work in other areas where authenticity and provenance are very important, such the markets for high-value art, luxury items, historical objects, or even managing the supply chain for consumer goods. It shows how blockchain could improve or replace traditional certificate-of-authenticity systems with ones that are more durable and can be checked by anybody. This study also helps people understand how blockchain can be used for more than just crypto-collectibles by showing that there is a viable alternative to NFT-only marketplaces. Instead of focusing on speculation, it focuses on real trust, transparency, and utility.

In the end, the OneFan project was able to make a safe and reliable marketplace for authenticated sports collectibles. This directly solved a big problem with authenticity in this field and showed how well blockchain technology and digital commerce can work together. Throughout the project, strategic design choices and proactive risk management helped the platform seamlessly integrate blockchain while still being easy to use and safe. For example, they abstracted the complexity of blockchain away from end users, used Solana's high-performance network for on-chain verification, and put in place strict security and privacy protections. Using Solana's blockchain to check the authenticity of non-NFTs worked well to make transactions more secure and keep users safe from fraud. The method that came out of this not only helps sports memorabilia fans right away, but it also shows how e-commerce can be improved through new ideas. As blockchain technology improve, projects like OneFan help make marketplaces more secure, open, and decentralized. This gives users more power and changes the way we build confidence in online transactions. The work done in this thesis lays a strong platform for future advances at the interface of blockchain and commerce, where trust and authenticity are key.

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